- **limited variety** of goods and services;
- **limited freedom of choice** for consumers and producers.
- Informational, administrative challenges; incentives for producing ≠ actual demand; little liberty (directing labor and rationing output restricts choices); shortages and surpluses (administered prices do not clear markets)

1.3 Understanding the world by use of models

Model: a simplified representation of something in the real world; it represents only the important aspects of the real world being investigated, ignoring unnecessary details.

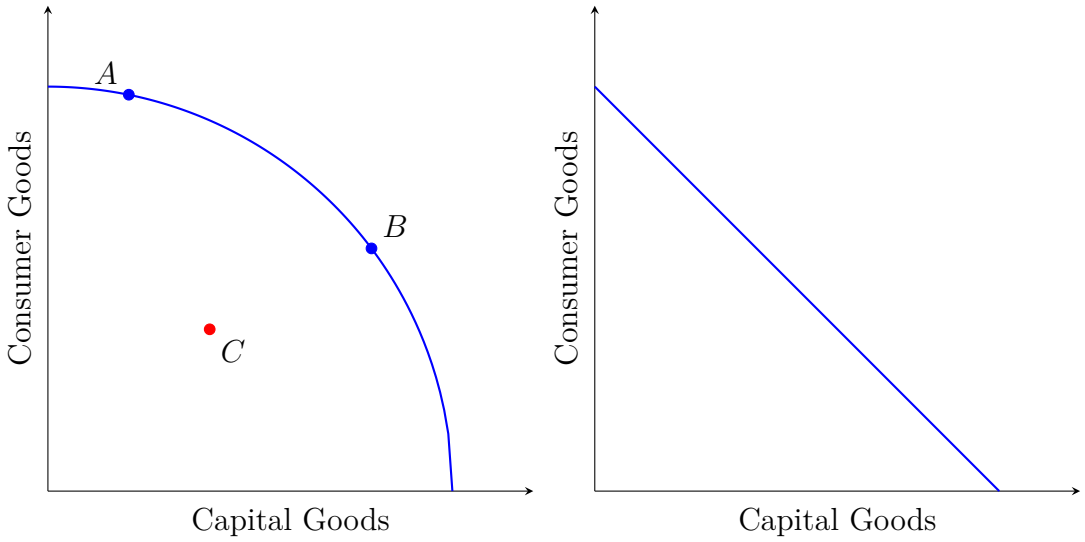
Production Possibility Curve / Frontier

Definition: the **production possibility curve (PPC)** represents all combinations of the maximum amounts of two goods that can be produced by an economy, given its resources and technology, when there is full employment of resources and efficiency in production.
All points on the curve are known as production possibilities.

Assumptions:

1. There is a fixed amount of resources
2. Only two goods are being produced
3. Technology and production techniques are fixed
4. All resources are used efficiently & fully employed

An economy's actual output is always at a point inside the PPC, because in the real world all economies have some unemployment of resources and some inefficiency in production.



Left: increasing OPC (concave PPC) — This happens because of specialisation of factors of production, which makes them not equally suitable for the production of different goods and services.
Right: constant OPC (straight-line PPC) — the factors of production are equally well suited to the production of both goods. Marginal rate of transformation (MRT) is constant throughout the curve.

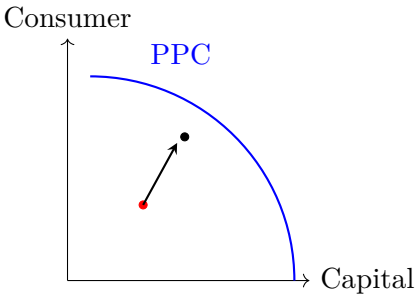
- Because of scarcity, the economy cannot produce outside its PPC. (其实是可以的，但这个要跨国交易了，所以暂不讨论)

- Because of scarcity, the economy must make a choice about what particular combination of goods will be produced.
- Because of scarcity, choices involve opportunity costs.

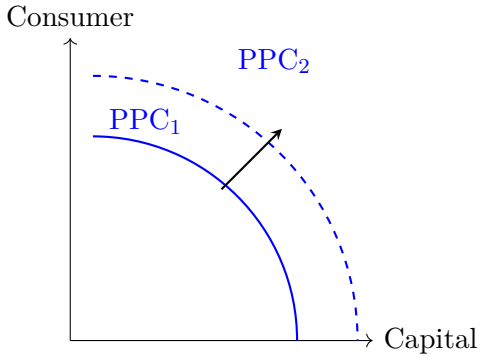
An economy's actual output, or the quantity of output actually produced, is always at a point inside the PPC, because in the real world all economies have some unemployment of resources and some inefficiency in production. The greater the unemployment or the inefficiency, the further away is the point of production from the PPC.

Economic growth: increases in the quantity of output produced in an economy over a period of time.
Causes of growth:

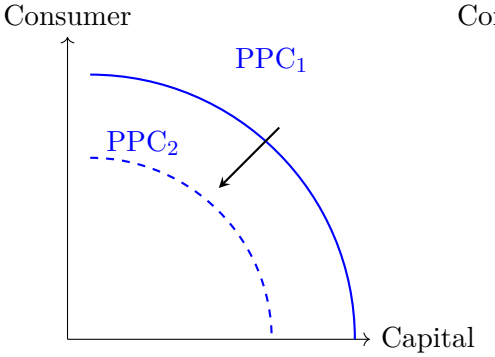
- Actual growth: reducing unemployment and increasing efficiency, point moves closer to PPC
- Growth in production possibilities:
  - increases in the quantity of resources / FoPs in the economy
  - improvements in the quality of resources
  - technological improvements



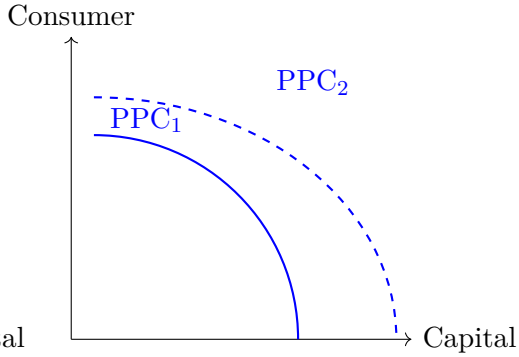
(a) Actual growth



(b) Potential growth



(c) Decrease



(d) Non-parallel shift

Circular Flow of Income Model

This model shows the interaction between households and firms.

- The model illustrates a closed economy, meaning it has no links with other countries, and is also a model of an economy with no government and no banks or financial market.
- It is assumed that the only decision-makers are households and firms.

Unit 2 Microeconomics

2 Competitive markets: Demand and supply

2.1 Introduction to competitive markets

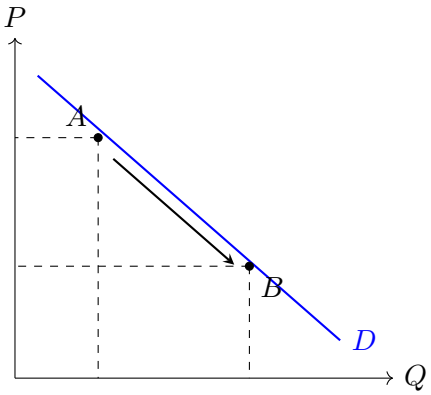
Markets: buyers and sellers of goods, services or resources are linked together to carry out an exchange. Goods and services are sold in product markets, while resources are sold in resource / factor markets.

Competition: process in which rivals compete in order to achieve some objective.
Competitive market: composed of large numbers of sellers and buyers acting independently, so that no one individual seller or small group of sellers has the ability to control the price of the product sold. (market / monopoly power) (through forces of demand and supply instead) Everyone is a price taker.

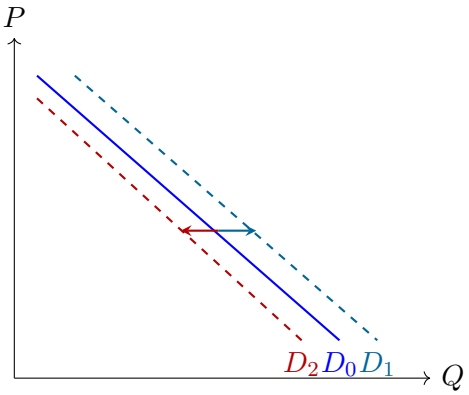
A market is any arrangement where those who are willing to and able to buy a good, service, or resource meet those who are willing and able to sell it, and a price is agreed.

2.2 Demand

Demand is concerned with the behaviour of buyers. (consumers generally, but can be applied to firms in resource market as well)
The demand of an individual consumer indicates the various quantities of a G&S the consumer is willing (wants to buy the good) and able (can afford to buy it) to buy at different possible prices during a particular time period, *ceteris paribus*.



Movement along the curve



Shifts of demand

Market demand is the sum of all individual demands for a good. The market demand curve illustrates the law of demand, shown by the negative relationship between price and quantity demanded.

The law of demand

There is a negative relationship between the price of a good and its quantity demanded over a particular time period, *ceteris paribus*: as the price of the good increases, quantity demanded falls; as the price falls, quantity demanded increases, *ceteris paribus*. This is reflected as the negative slope of the demand curve. The higher the price of a good, the less of it you are probably willing and able to buy.

Non-price determinants of demand

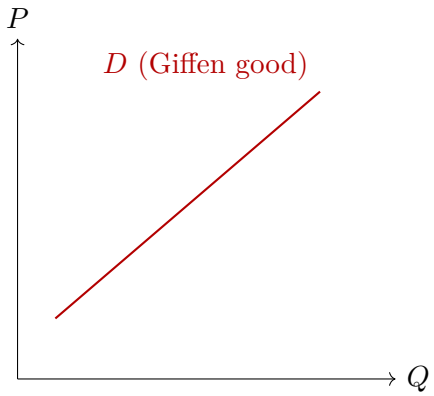
- Income in the case of normal goods (demand for the good varies directly with income)
  - An increase in income leads to a rightward shift in the demand curve.

According to the **law of diminishing marginal utility**, as consumption of a good increases, marginal utility, or the extra utility the consumer receives, decreases with each additional unit consumed. This underlies the law of demand, as it shows that a consumer will be willing to buy an additional unit of a good only if its price falls.

Income and substitution effect

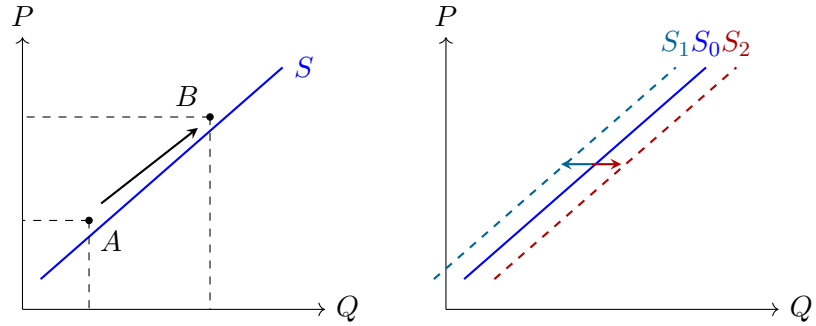
- **The substitution effect**
  - If the price of a good falls, the consumer substitutes (buys more) of the now less expensive good. Therefore quantity demanded increases.
- **The income effect**
  - If the real income (purchasing power) increases and price falls, quantity demanded of the good increases.

In the case of most goods, whose price is a small fraction of income, the substitution effect is the most important part of the explanation of the law of demand, because being a small fraction of income, the effect that a price change has on real income will be tiny. The income effect becomes important only in the case of purchases of goods whose price take up a large fraction of income. The above analysis refers to **normal goods**. In the case of **inferior goods**, as price falls and real income increases, consumers switch to better alternatives, so the **income effect is negative** (works *against* the substitution effect). For most inferior goods, the substitution effect still *dominates* the income effect, so the demand curve remains downward sloping.
Very rarely, the negative income effect may **exceed** the substitution effect, so that a fall in price leads to a *decrease* in quantity demanded. This produces an **upward-sloping demand curve**, and the good is called a **Giffen good** — extremely rare in practice.



2.3 Supply

Supply is concerned with the behaviour of sellers, which include firms in the product markets and households in resource markets.
The supply of an individual firm indicates the various quantities of a G&S a firm is willing and able to produce and supply to the market for sale at different possible prices, during a particular time period, *ceteris paribus*.
Market supply indicates the total quantities of a good that firms are willing and able to supply in the market at different possible prices and is given by the sum of all individual supplies of that good.
As with demand, the firm's supply can be presented as a **supply schedule** (a table listing quantity supplied at various prices), which is then plotted as the **supply curve**. Market supply is the horizontal sum of all individual firms' supply schedules.



The law of supply

There is a positive relationship between the quantity of a good supplied over a particular time period and its price, *ceteris paribus*: as the price of the good increases, the quantity of the good supplied also increases; as the price falls, the quantity supplied also falls, *ceteris paribus*. (As price increases, there's more incentive for producers to produce due to increasing profit earned)

The vertical supply curve

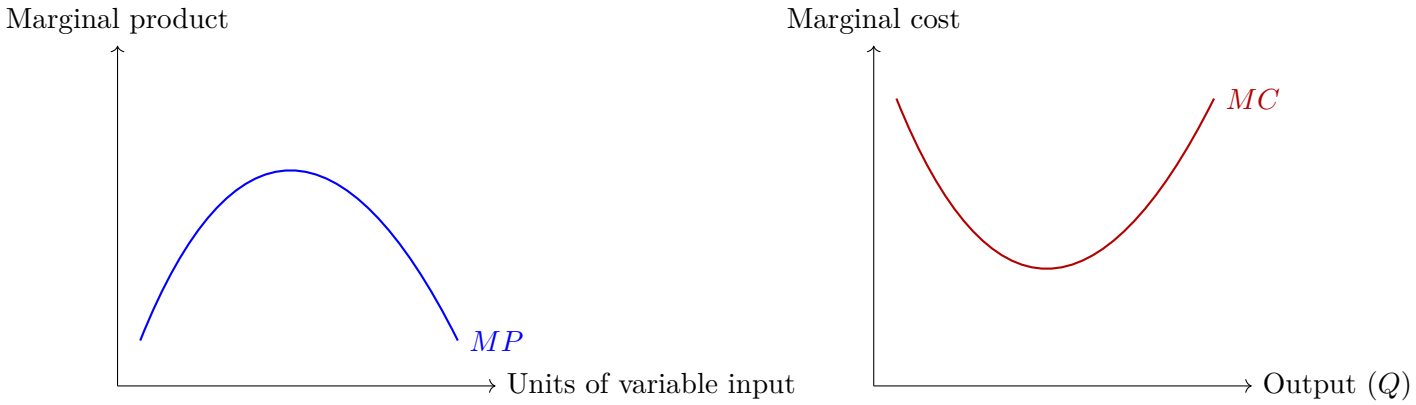
A vertical supply curve tells us that even as price increases, the quantity supplied cannot increase; it remains constant. The quantity supplied is independent of price. There are two reasons why this may occur:

- There is a fixed quantity of the good supplied because there is no time to produce more of it.
- There is a fixed quantity of the good because there is no possibility of ever producing more of it.

Non-price determinants of supply curve

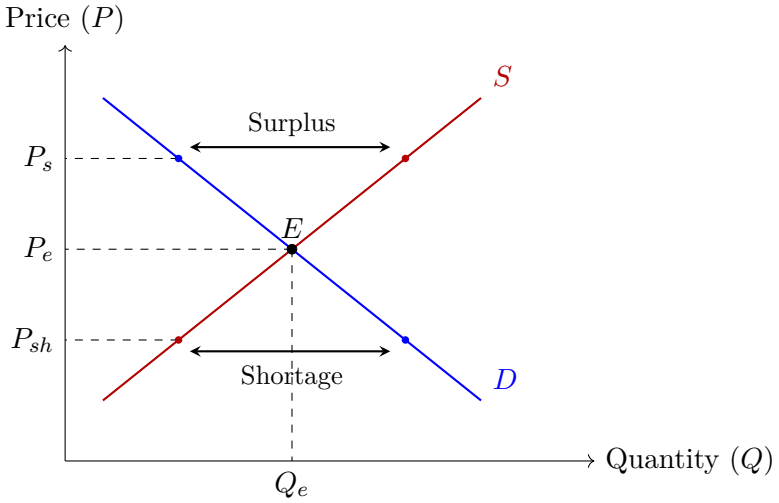
- Costs of resources / FoPs
- Technological improvements
- Price of related goods: Competitive supply
  - Competitive supply of two or more products refers to production of one or the other by a firm.
  - Resources are often limited, therefore producers must choose which type of product should the resource be devoted to.
- Prices of related goods: Joint supply
  - Joint supply of two or more products refers to production of goods that are derived from a single product, so that it is not possible to produce more of one without producing more of the other.
- Producer price expectation
  - If firms expect the price of their product to rise, they may withhold some of their current supply from the market, expecting that they will be able to sell it at the higher price in the future.
- Taxes (indirect / taxes on profits) (treated as increase in CoP)
- Subsidies: a payment made to the firm by the government, = decrease in CoP
- Size of the market / the number of firms (increase leads to rightward shift of supply curve)
- 'Shocks' or sudden unpredictable events

to cover its costs. Therefore we can think of the supply curve as showing the price that the firm is willing to accept to produce one more unit of the good. The supply curve begins at the point on the marginal cost curve where the firm is making enough revenue so that it is better off producing than shutting down. (P>AVC)



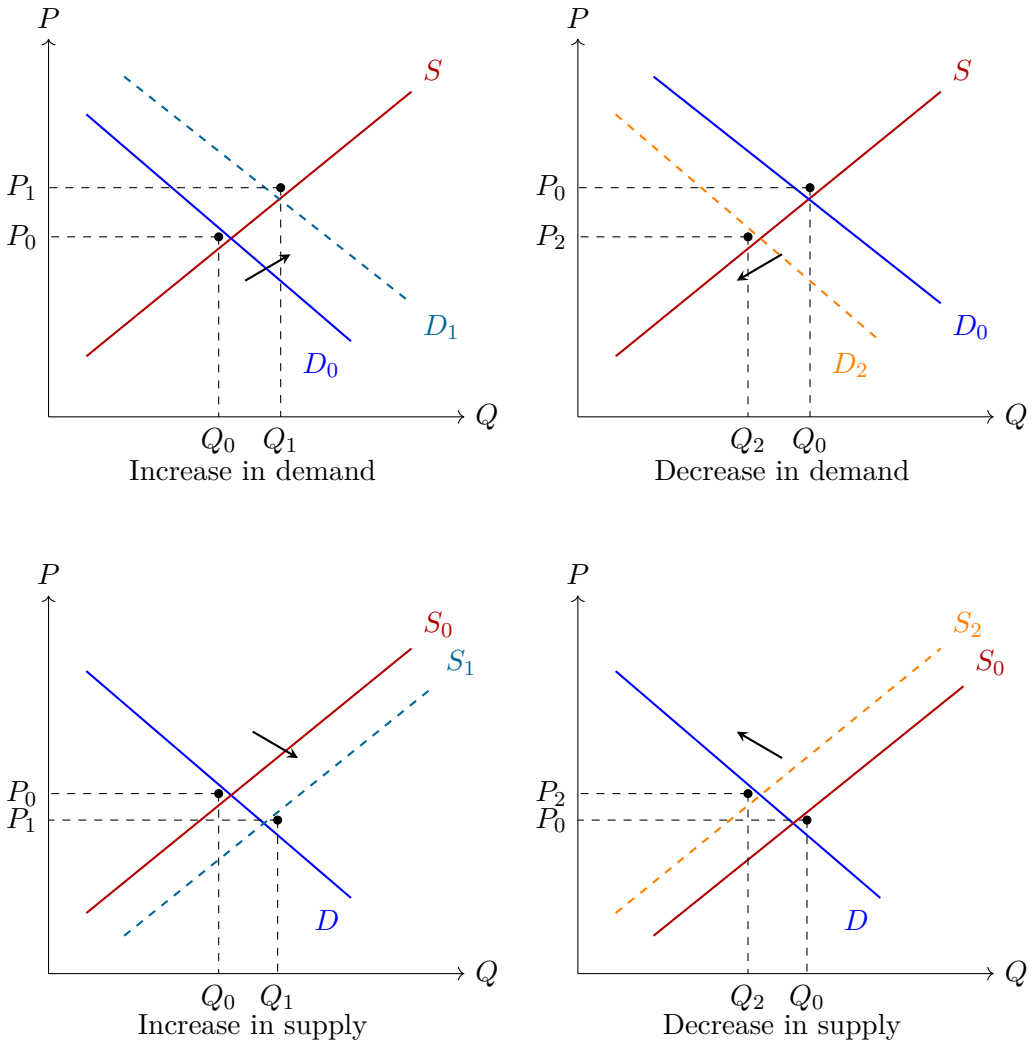
2.4 Competitive market equilibrium

Shortages: excess demand. Surplus: excess supply.
The existence of excess demand (a shortage) or excess supply (a surplus) in a free market will cause the price to change so that the quantity demanded will be made equal to quantity supplied. In the event of excess demand, price will rise; in the event of excess supply, price will fall.、、、、 Market equilibrium occurs at a price where quantity demanded = quantity supplied; no surplus / shortage.



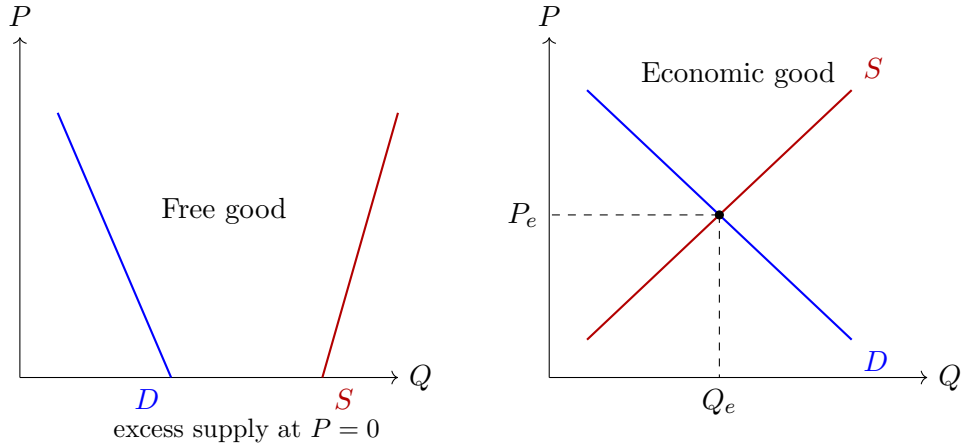
When quantity demanded is equal to quantity supplied, there is market equilibrium; the forces of supply and demand are in balance, and there is no tendency for the price to change.

Changes in market equilibrium



Free goods revisited

A free good is a good for which the quantity supplied is greater than the quantity demanded when the price is zero. Supply is so large relative to demand that there is excess quantity supplied even at a zero price. A free good can change into an economic good as a result of a leftward shift in the supply curve (reduced supply) or a rightward shift in the demand curve (increased demand). When demand and supply intersect at a price greater than zero, the good has become an economic good.



In order to achieve allocative efficiency, $MB = MC$. Practically, it means that the extra benefit to society of getting one more unit of the good is equal to the extra cost to society of producing one more unit of the good. When this happens, society's resources are being used to produce the 'right' quantity of the good.

- If $MB > MC$, society would be placing a greater value on the last unit of the good produced than it costs to produce it, and so more of it should be produced.
- If $MB < MC$, it would be costing society more to produce the last unit of the good produced than the value society puts on it, and so less should be produced.

Consumer and producer surplus

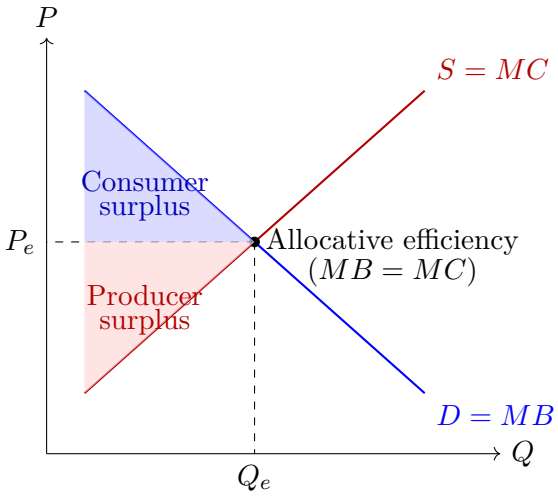
Consumer surplus (for a specific quantity) is defined as the price consumers are willing to pay for a good minus the price actually paid. It is the area under the demand curve and below the price paid by the consumer, up to the quantity purchased.

$$\text{Consumer Surplus} = \frac{1}{2}(P_{\text{intercept of D}} - P_e) \times Q_e$$

Producer surplus (for a specific quantity) is defined as the price received by firms for selling their good minus the price that they are willing to accept to produce the good. It is shown as the area above the firms' supply curve and below the price received by the firm, up to the quantity produced.

$$\text{Producer Surplus} = \frac{1}{2}(P_e - P_{\text{intercept of S}}) \times Q_e \left(= \frac{1}{2}(Q_{\text{intercept of S}} + Q_e) \times P_e \right)$$

(Note when supply curve has a Q-intercept, producer can be determined using formula for area of a trapezium)



At the point of competitive market equilibrium, the sum of consumer and producer surplus is maximum. At equilibrium, social surplus, defined as the sum of consumer plus producer surplus, is maximum. Therefore maximum social surplus is just another way of saying that there is allocative efficiency. Social welfare, the amount of consumer and producer surplus, is maximized as well.

Allocative efficiency of markets vs Govt. Intervention

Issues with market allocative efficiency:

- Efficiency can only arise under a number of very strict and highly unrealistic conditions that are practically never met in the real world.

3.1 Price elasticity of demand (PED)

Price elasticity of demand is a measure of the responsiveness of the quantity of a good demanded to changes in its price. (Note that although the value calculated is going to be negative, the common practice is to drop the minus sign and consider PED as a positive number / take the absolute value)

- If quantity demanded is highly responsive to a change in price, $PED > 1$ demand is price elastic; change in price leads to a proportionally larger change in Q_d
- If quantity demanded is not very responsive, $PED < 1$ demand is price inelastic; change in price leads to a proportionally smaller change in Q_d
- Demand is unit elastic when $PED = 1$, also called unitary PED.
- Demand is perfectly inelastic when $PED = 0$.
- Demand is perfectly elastic when $PED = \infty$.

$$PED = \frac{\%\Delta Q_d}{\%\Delta P} = \frac{\dfrac{Q_2 - Q_1}{Q_1}}{\dfrac{P_2 - P_1}{P_1}}$$

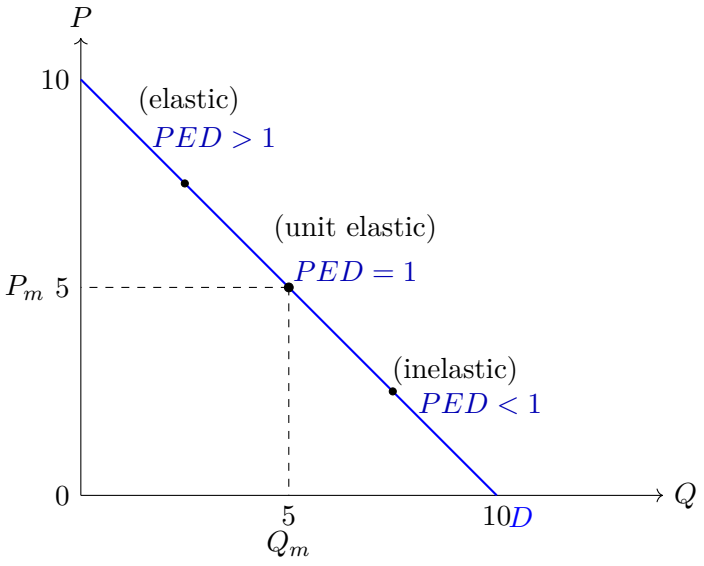
Midpoint formula for PED calculation:

$$PED = \frac{\dfrac{\Delta Q_d}{Q_{\text{avg}}}}{\dfrac{\Delta P_d}{P_{\text{avg}}}} = \frac{\dfrac{\Delta Q}{\frac{1}{2}(Q_1 + Q_2)}}{\dfrac{\Delta P}{\frac{1}{2}(P_1 + P_2)}}$$

In general, when demand curves intersect, then for any given price, the flatter the demand curve, the more elastic is the demand. The steeper the demand curve, the less elastic the demand.

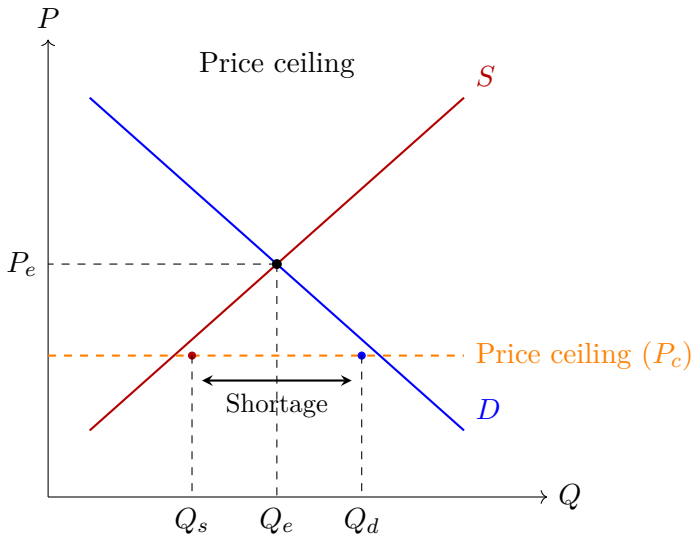
Changing PED and the straight-line demand curve

Along any downward-sloping, straight-line demand curve, the PED varies (changes) as we move along the curve.



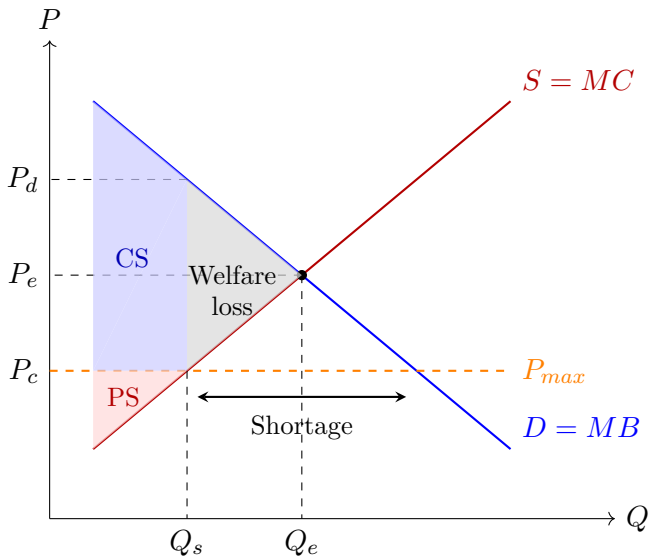
4.2 Price controls

Price controls refer to the setting of minimum or maximum prices by the government (or private organisations) so that prices are unable to adjust to their equilibrium level determined by demand and supply. Price controls result in market disequilibrium, and therefore in shortages (excess demand) or surpluses (excess supply).



A **price ceiling** is a maximum price set below the equilibrium price, in order to make goods more affordable to people on low incomes. Examples: rent and food controls. Consequences of price ceilings:

- For markets
  - Creation of shortage $= Q_d - Q_s$
  - Non-price rationing in order to distribute Q_s among all interested buyers
  - Underground / parallel markets (buying a good at legal max price, sell it at price higher than that)
  - Underallocation of resources to the good (underproduction, $Q_s < Q_e$) and allocative inefficiency
  - Negative welfare impacts: Welfare loss (also known as deadweight loss) represents social surplus or welfare benefits that are lost to society because resources are not allocated efficiently.



At the price ceiling P_{max}, only Q_s is supplied (and transacted). **Consumer surplus (CS)** is the area above P_{max} and below the $D = MB$ curve up to Q_s. **Producer surplus (PS)** is the area

below P_{max} and above the $S = MC$ curve up to Q_s. The shaded triangle between Q_s and Q_e is the **welfare loss (deadweight loss)** — output that would have benefited society ($MB > MC$) is not produced due to the price ceiling.

- For stakeholders
  - Consumers: some consumers are able to buy the good at the lower price are better off. However, some consumers remain unsatisfied since they don't get to buy the good at all.
  - Producers: worse off, they sell a lower quantity of goods at a lower price, resulting in decreased revenue.
  - Workers: resulting in unemployment

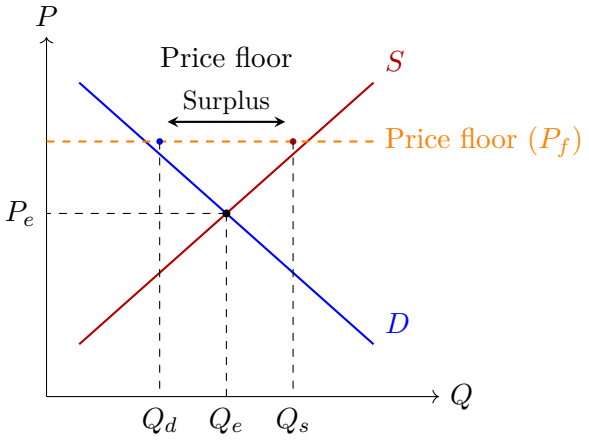
Calculating the effects of price ceilings on stakeholders and welfare

Change in consumer spending / firm revenue: $P_e \times Q_e - P_c \times Q_s$.
Change in consumer and producer welfare and welfare loss:

$$\Delta\text{Consumer surplus} = \frac{1}{2}Q_e \times (P_{\text{intercept of D}} - P_e) - \frac{1}{2}Q_s \times ((P_d - P_c) + (P_{\text{intercept of D}} - P_c))$$

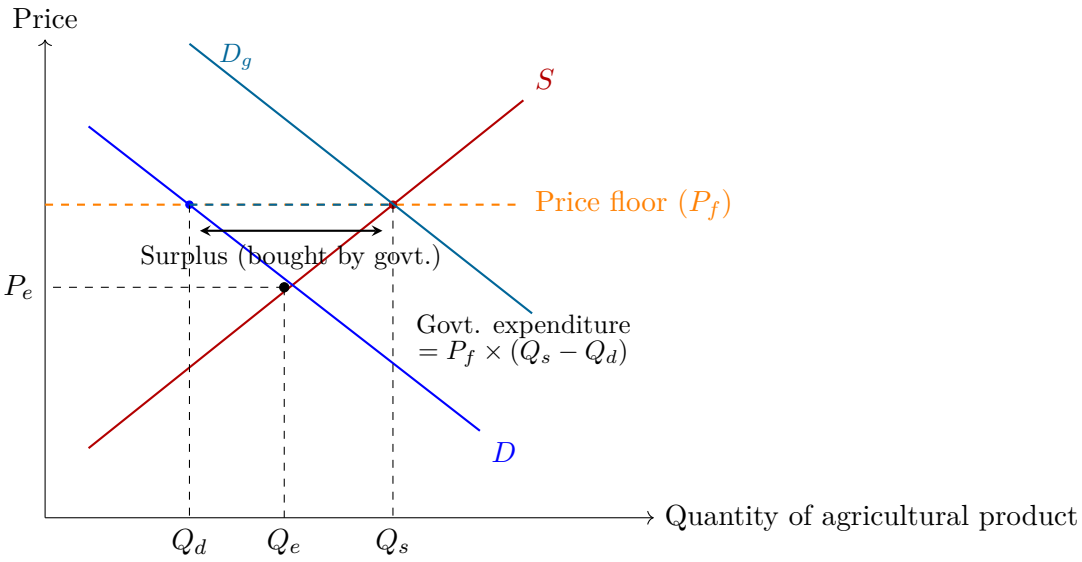
$$\Delta\text{Producer surplus} = \frac{1}{2}Q_e \times (P_e - P_{\text{intercept of S}}) - \frac{1}{2}(P_e - P_c) \times (Q_s + Q_e)$$

$$\text{Welfare loss} = -(\Delta\text{Consumer surplus} + \Delta\text{Producer surplus}) = \frac{1}{2}(P_d - P_c) \times (Q_e - Q_s)$$

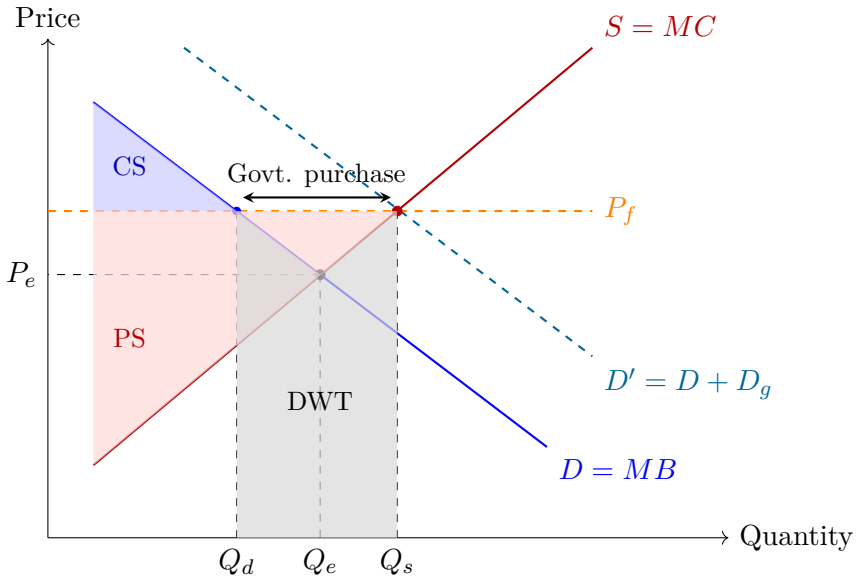


A **price floor** is a minimum price set above the equilibrium price, in order to provide income support to farmers or to increase the wages of low-skilled workers. Consequences of price floors:

- For markets
  - Creation of surplus: $Q_s - Q_d$
  - A common practice is for the government to buy the excess supply, and this causes the demand curve for the product to shift to the right.



- The government must make a decision about what to do with the surplus: store / export?
- Firm inefficiency: inefficient firms with high costs of production do not face incentives to cut costs because the high price offers them protection against lower-cost competitors.
- Overallocation of resources to the production of the good and allocative inefficiency
- Negative welfare impacts:



Government spending to maintain the price floor involves losses for society. The grey shaded area is the DWT (while the producers gain the additional red triangle, the government pays a price of the grey rectangle) $MB < MC$, indicating that society would be better off if less of the good were produced.

- For stakeholders
  - Consumers: worse off, pay a higher price for the good
  - Producers: better, gain as they receive a higher price and produce a larger quantity
  - Workers: employment increases
  - Government: a burden on its budget, also incur OPC

Calculating effects of price floors on stakeholders and social welfare

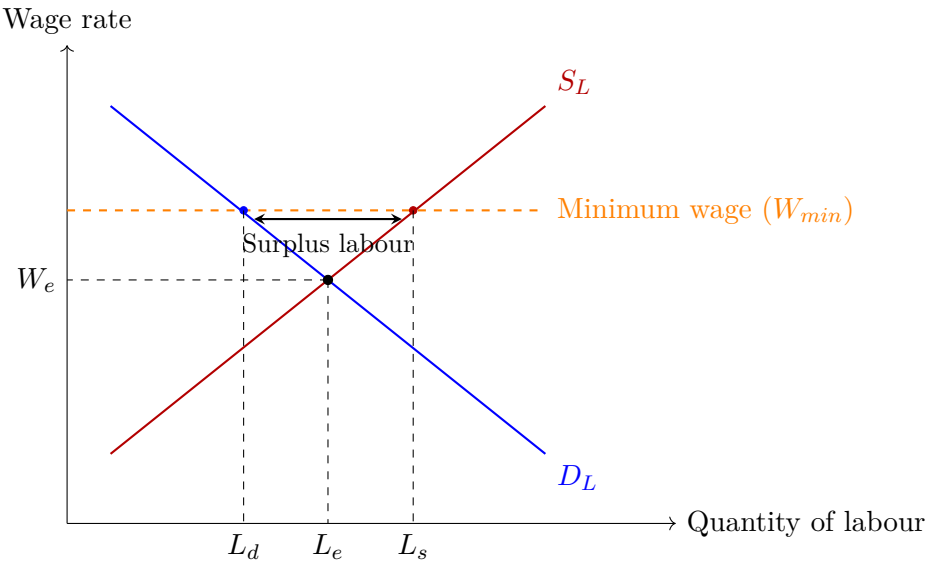
Changes in consumer expenditure: $P_f \times Q_d - P_e \times Q_e$
Changes in producer revenue: $P_f \times Q_s - P_e \times Q_e$
Changes in government expenditure: $P_f \times (Q_s - Q_d)$

$$\Delta\text{Consumer surplus} = \frac{1}{2}(Q_d + Q_e) \times (P_f - P_e)$$

$$\Delta\text{Producer surplus} = \frac{1}{2}(Q_e + Q_s) \times (P_f - P_e)$$

$$\Delta\text{Welfare loss} = (Q_s - Q_d) \times P_f - \frac{1}{2}(Q_s - Q_d) \times (P_f - P_e)$$

Minimum wages



A **minimum wage** (W_{min}) is a price floor in the labour market, set above the equilibrium wage W_e. Consequences of minimum wages:

- Labour surplus (excess supply) and unemployment
- Illegal workers at wages below the minimum wage
- Misallocation of labour resources
- Misallocation in product markets

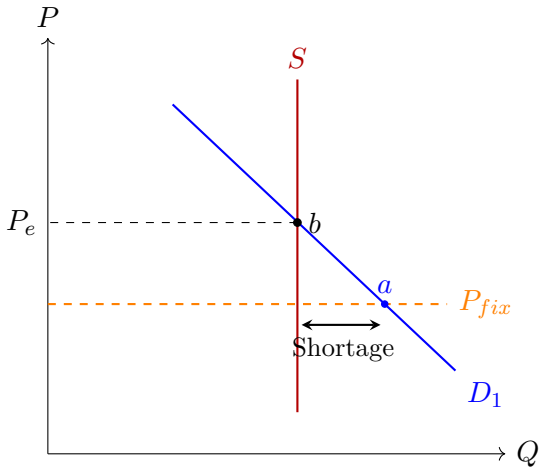
Consequences for various stakeholders:

- Firms: worse off, increase CoP
- Workers: some receive higher wages, some experience unemployment
- Consumers: extra CoP reflected in higher prices

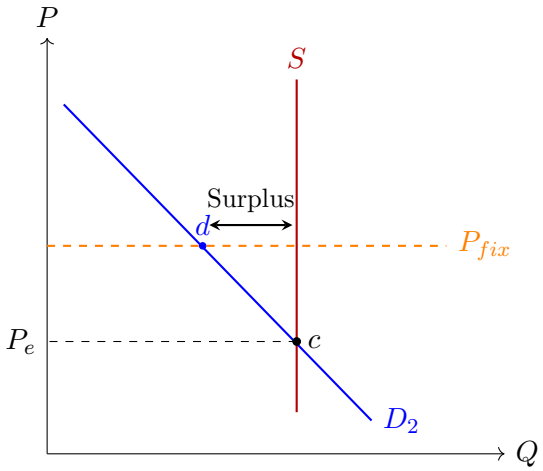
The level of unemployment created largely depends on the minimum wage set. It may even increase labor productivity due to an increase in incentive for work.

Setting fixed prices

Sometimes prices may be fixed at a particular level by the government (e.g. tickets for a sports event with a fixed number of seats). The supply curve is **vertical**. If the fixed price is set below the equilibrium, it creates a **shortage**; if set above, it creates a **surplus**.



(a) High demand: shortage



(b) Low demand: surplus

4.3 Indirect taxes

Indirect taxes are imposed on spending to buy goods and services. Types of indirect tax include:

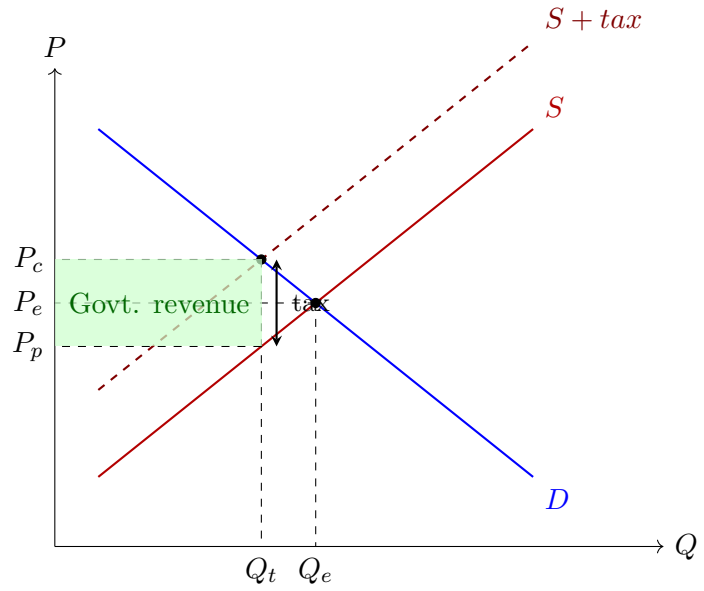
- excise taxes, imposed on particular goods and services, such as petrol, cigarettes and alcohol
- taxes on spending on all (or most) goods and services, such as general sales taxes and value added taxes.

Why do governments impose excise taxes?

- Indirect taxes are a source of government revenue.
- Indirect taxes are a method to discourage consumption of goods that are harmful for the individual.
- Indirect taxes can be used to redistribute income.
- Indirect taxes are a method to improve the allocation of resources (reduce allocative inefficiencies) by correcting negative externalities.

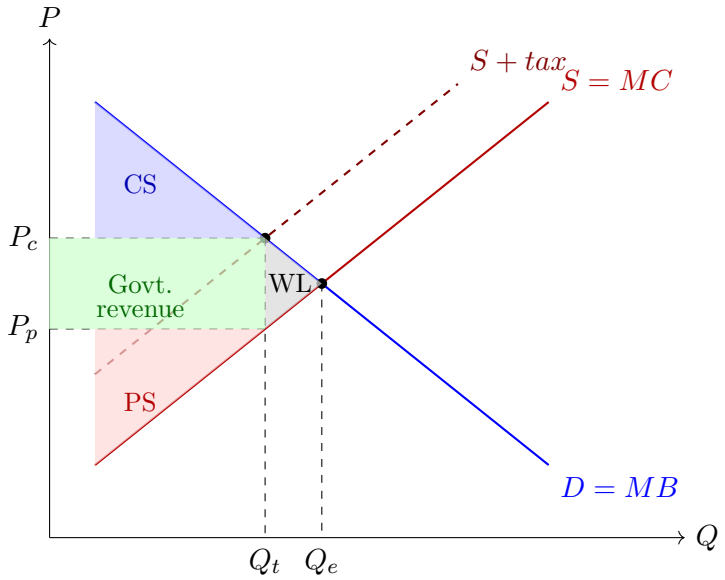
Indirect, excise taxes can be:

- **specific taxes**, a fixed amount of tax per unit of the good or service sold;
- **ad valorem taxes**, a fixed percentage of the price of the good or service.



An indirect (excise) tax shifts the supply curve **upward** from S to $S+tax$. The new equilibrium has a **lower quantity** $Q_t < Q_e$. Consumers pay a **higher price** P_c; producers receive a **lower price** P_p (after paying the tax). The green rectangle $= (P_c - P_p) \times Q_t$ is the **government tax revenue**. Consequences of indirect taxes for various stakeholders:

- Consumers: Increase in price, decrease in quantity bought
- Firms: fall of output and fall of revenue they receive (decreased price)
- Governments: Gains revenue
- Workers: leads to some unemployment
- Society as a whole: CS, PS, DWT
  In this case, welfare loss appears because the tax causes a smaller than optimum quantity to be produced, which is an indication of underproduction.



Calculating the effects of indirect taxes on stakeholders and social welfare

Change in consumer expenditure: $P_e \times Q_e - P_c \times Q_t$
Change in producer revenue: $P_e \times Q_e - P_p \times Q_t$

Government revenue: $(P_c - P_p) \times Q_t$

Change in CS: $\frac{1}{2}(P_{\text{intercept of D}} - P_e) \times Q_e - \frac{1}{2}(P_{\text{intercept of D}} - P_c) \times Q_t$

Change in PS: $\frac{1}{2}(P_e - P_{\text{intercept of S}}) \times Q_e - \frac{1}{2}(P_p - P_{\text{intercept of S}}) \times Q_t$

DWT / welfare loss: $\frac{1}{2}(P_c - P_p) \times (Q_e - Q_t)$

Tax incidence and price elasticities of demand and supply

Tax incidence refers to the distribution of the burden of a tax between consumers and producers. When a good is taxed, consumers pay a higher price ($P_c > P_e$) and producers receive a lower price ($P_p < P_e$). The total tax revenue $= (P_c - P_p) \times Q_t$ is shared between the two groups:

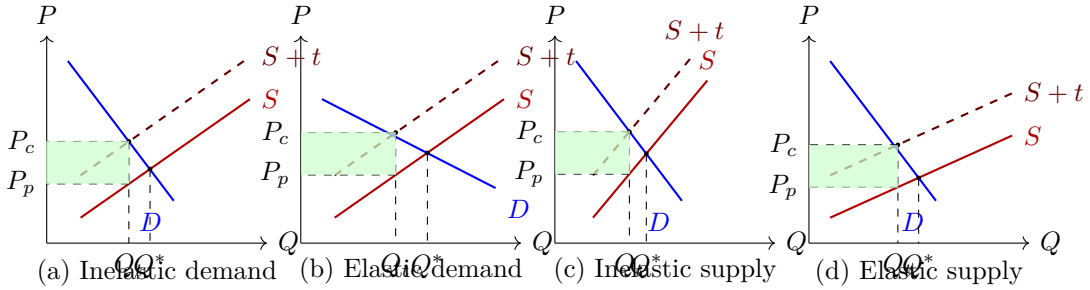
- **Tax burden on consumers** $= (P_c - P_e) \times Q_t$
- **Tax burden on producers** $= (P_e - P_p) \times Q_t$

The distribution depends on **PED and PES**:

- **Inelastic demand** (steep D): consumers are unresponsive to price changes, so they bear **most** of the tax burden. The drop in quantity ($Q_e \to Q_t$) is small.
- **Elastic demand** (flat D): consumers are responsive and can easily switch to substitutes, so producers bear **most** of the tax burden. The drop in quantity is larger.

Similarly for **PES**:

- **Inelastic supply** (steep S): producers bear most of the tax burden.
- **Elastic supply** (flat S): consumers bear most of the tax burden.



Summary rule: the group with the more **inelastic** response (lower elasticity) pays the **higher share** of the tax. Therefore:

- If $PES > PED \to$ tax incidence mainly on **consumers**.
- If $PED > PES \to$ tax incidence mainly on **producers**.

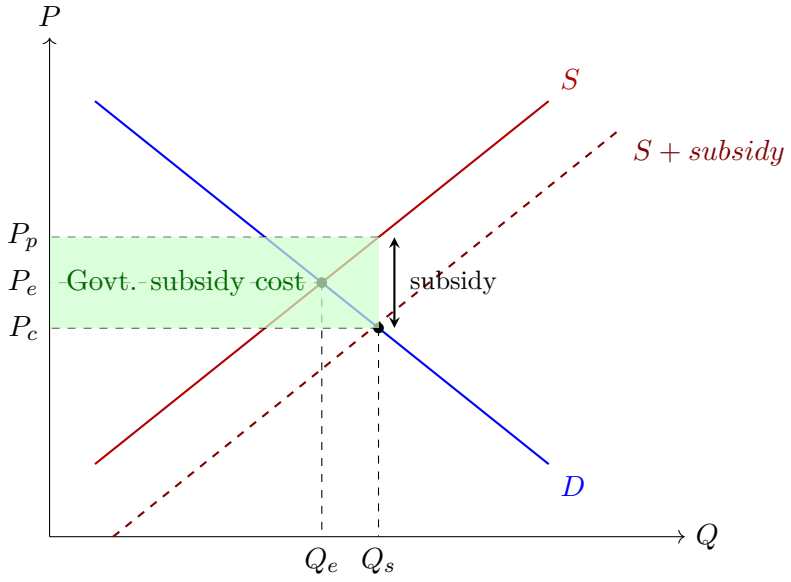
The less responsive group cannot easily adjust its behaviour and so absorbs more of the tax.

4.4 Subsidies

A **subsidy**, in a general sense, refers to assistance by the government to individuals or groups of individuals, such as firms, consumers, industries or sectors of an economy. A subsidy granted to a firm (or group of firms) has the effect of increasing the price received by producers, causing them to produce more, and lowering the price paid by consumers, causing them to buy more.

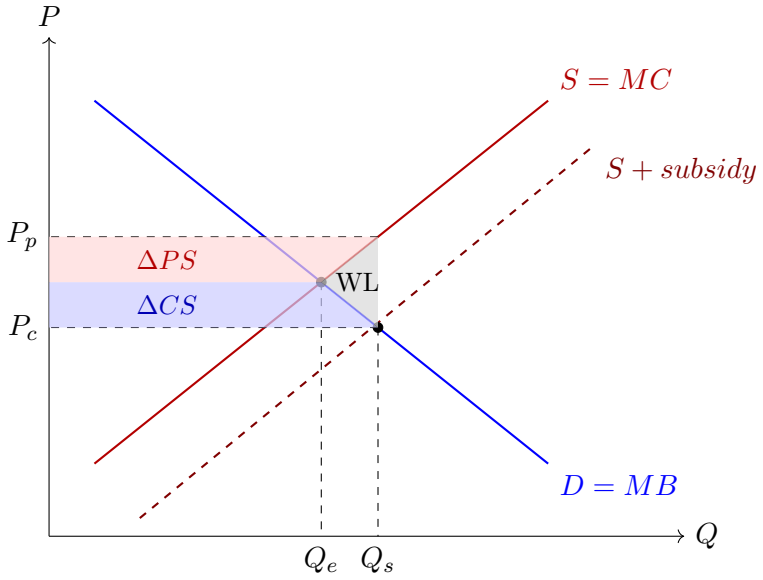
Why government grant subsidies

- Subsidies can be used to increase revenues (and hence incomes) of producers.
- Subsidies can be used to make certain goods (necessities) affordable to low-income consumers.
- Subsidies can be used to encourage production and consumption of particular goods and services that are believed to be desirable for consumers.
- Subsidies can be used to support the growth of particular industries in an economy.
- Subsidies can be used to encourage exports of particular goods.
- Subsidies area method to improve the allocation of resources (reduce allocative inefficiencies) by correcting positive externalities.



A subsidy shifts the supply curve **downward** (rightward) from S to $S + subsidy$. The equilibrium quantity **increases** from Q_e to Q_s. Consumers pay a **lower price** P_c; producers receive a **higher price** P_p (the government pays the difference). The green rectangle $= (P_p - P_c) \times Q_s$ is the total **cost of the subsidy to the government**. There is an overallocation of resources to the production of the good. Consequences for various stakeholders:

- Consumers: decrease in price, increase in quantity bought
- Producers: increase in quantity supplied, increase in revenue (receives P_p)
- Government: pays $(P_p - P_c) \times Q_s$ to the producer.
- Workers: increased employment
- Society as a whole:



The granting of a subsidy results in greater consumer and producer surplus; however, society loses due to government spending on the subsidy. Since the loss from government spending is greater than the gain in consumer and producer surplus, welfare loss results, reflecting allocative inefficiency, which in this case is due to overallocation of resources to the production of the good (overproduction).

Calculating the effects of subsidies on market outcomes and social welfare

Change in consumer expenditure: $P_c \times Q_s - P_e \times Q_e$
Change in producer revenue: $P_p \times Q_s - P_e \times Q_e$
Government expenditure: $(P_p - P_c) \times Q_s$
Change in CS: $\frac{1}{2}(Q_e + Q_s) \times (P_e - P_c)$
Change in PS: $\frac{1}{2}(Q_e + Q_s) \times (P_p - P_c)$
DWT / welfare loss: $\frac{1}{2}(Q_s - Q_e) \times (P_p - P_c)$

4.5 Supplementary: Government failure (Edexcel Book 1 Ch 20)

Government failure occurs when government intervention creates a **worse** outcome than the market failure it was meant to correct (or causes a new inefficiency, leading to a net welfare loss). Causes:

- **Information failure / inadequate information**: governments cannot possess full knowledge of consumer preferences, true costs, or market dynamics, leading to wrong tax/subsidy levels or misallocation.
- **Regulatory capture**: regulatory bodies set up to protect public interest end up serving the commercial interests of the dominant firms they are supposed to regulate.
- **Policy myopia / short-termism**: politicians prioritise short-term electoral cycles over long-run economic efficiency (e.g. populist spending or tax cuts).
- **Unintended consequences**: policies frequently produce unanticipated side-effects (e.g. rent controls leading to severe housing shortages and black markets; generous unemployment benefits reducing work incentives).
- **Administrative and bureaucratic costs**: the enforcement, monitoring, and compliance costs of intervention may outweigh the social benefits generated.

- **Equity–efficiency trade-offs**: aggressive redistribution policies (high progressive taxes and transfer payments) can distort market price signals and reduce incentives to work and invest.

5 Market failure and socially undesirable outcomes I

5.1 The meaning of common pool resources

Common pool resources are resources that are not owned by anyone, do not have a price and are available for anyone to use without payment or any other restriction.

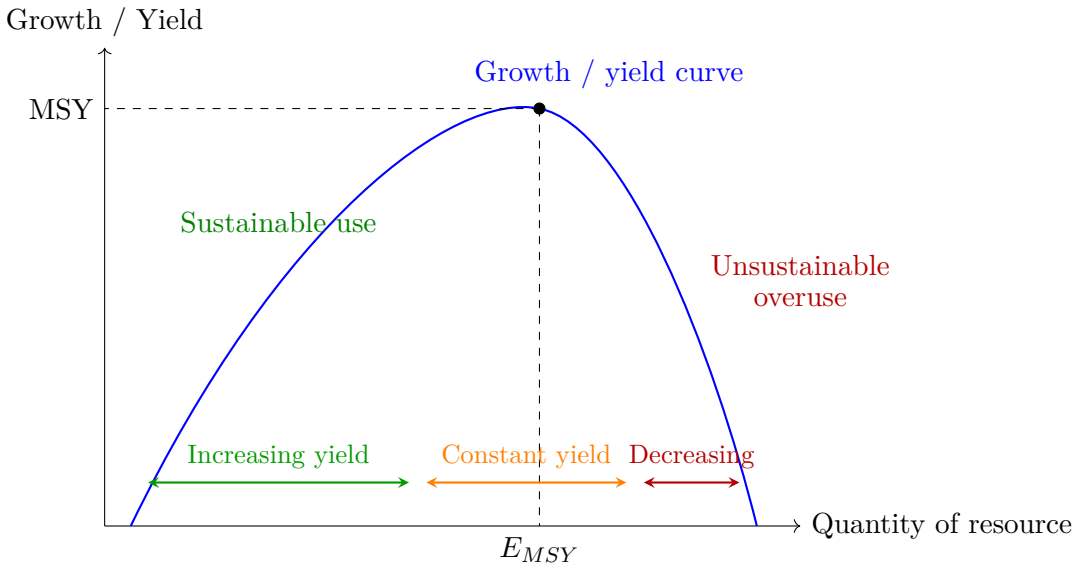
Two important concepts:

- **Rivalrous**: its consumption by one person reduces its availability for someone else.
- **Excludable**: it is possible to exclude people from using the good (by charging a price)

Common pool resources are rivalrous but nonexcludable.

Sustainability and common pool resources

Sustainable resource use means using a resource at a rate that allows it to regenerate, so its quantity and quality are maintained over time. For biological resources (forests, fish stocks), the key concept is the **maximum sustainable yield (MSY)** — the largest quantity that can be harvested indefinitely without depleting the resource.



- **Increasing yield phase** (low effort): as harvesting effort increases, yield increases. The resource is being used below its regenerative capacity (**underuse**).
- **Maximum sustainable yield (MSY)**: the peak of the growth curve. Harvesting at E_{MSY} allows the resource to regenerate indefinitely — this is **sustainable use**.
- **Decreasing yield phase** (effort beyond E_{MSY}): over-harvesting reduces the resource's ability to reproduce, so yield falls even as effort increases. Eventually the resource is depleted (**unsustainable overuse**).

Sustainable resource use means that resources are used at a rate that allows them to reproduce themselves, so that they do not become degraded or depleted.

5.2 Market failure and externalities

Market failure refers to the failure of the market to allocate resources efficiently. Market failure results in allocative inefficiency, where too much or too little of goods or services are produced and consumed from the point of view of what is socially most desirable.

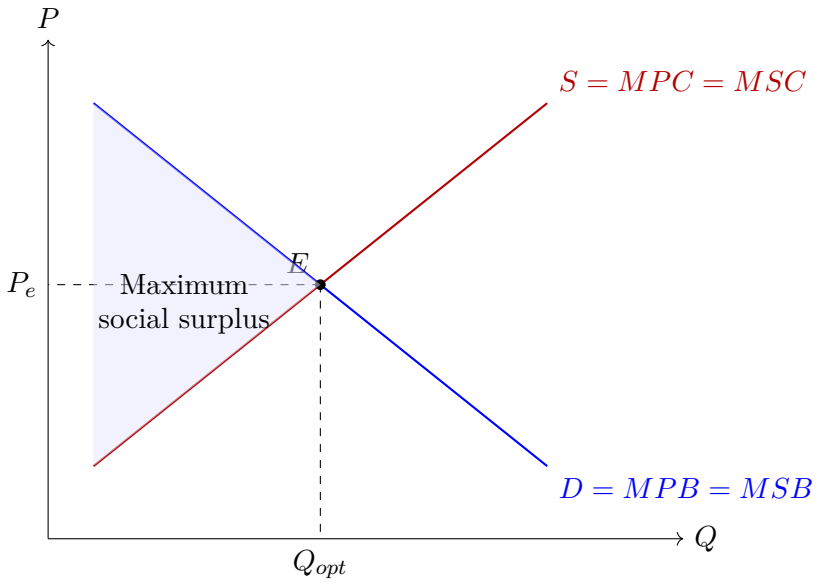
- Overprovision of a good means too many resources are allocated to its production (overallocation);
- Underprovision means that too few resources are allocated to its production (underallocation).

An **externality** occurs when the actions of consumers or producers give rise to negative or positive side-effects on other people who are not part of these actions, and whose interests are not taken into consideration.

- If the side-effects on third parties involve benefits, there arises a positive externality, also known as external (or spillover) benefit;
- If they involve costs, in the form of negative side-effects, there arises a negative externality, also known as external (or spillover) costs.

Externalities can result either from consumption activities or production activities. When there are **no externalities**, private costs/benefits and social costs/benefits coincide:

- **Demand** $= MPB = MSB$ (marginal private benefit = marginal social benefit)
- **Supply** $= MPC = MSC$ (marginal private cost = marginal social cost)



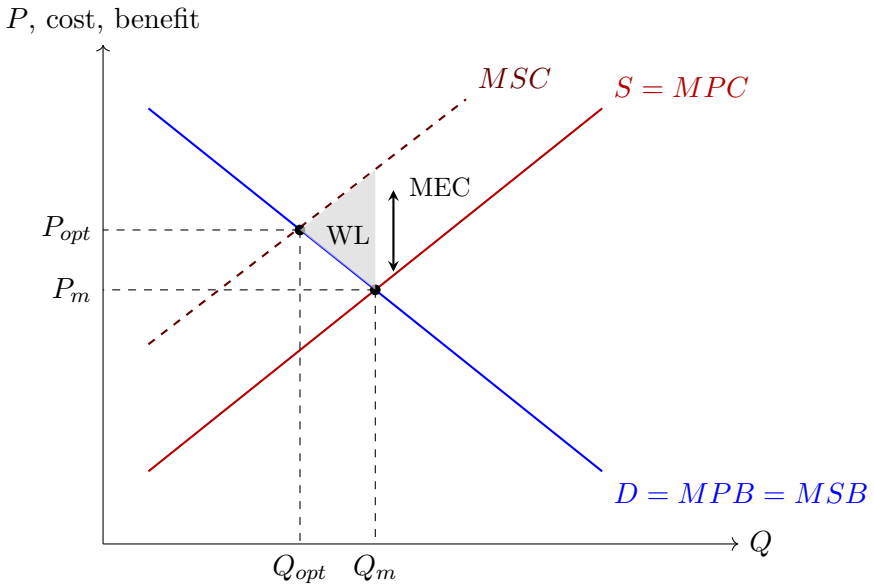
With no externalities, the free-market equilibrium at Q_{opt} achieves **allocative efficiency**: $MSB = MSC$, social surplus is maximised, and there is no welfare loss. The socially optimal quantity Q_{opt} is identical to the market equilibrium quantity.
If, however, there is an externality, additional benefits or additional costs affecting third parties arise, and the full benefits or full costs to society differ from the private ones. When this occurs, the equilibrium price and quantity determined by the intersection of the demand (MPB) curve and supply (MPC) curve is no longer a social optimum.

- MPC: costs to producers of producing one more unit of a good.
- MSC: costs to society of producing one more unit of a good.
- MPB: benefits to consumers from consuming one more unit of a good.
- MSB: benefits to society from consuming one more unit of a good.

5.3 Negative production externalities

Negative production externalities occur when production imposes **external costs** on third parties (e.g. pollution from a factory). The social cost of production exceeds the private cost:

- $MSC = MPC + MEC$ (marginal external cost)
- Since $MEC > 0$, $MSC > MPC \Rightarrow$ the MSC curve lies **above** the MPC (S) curve.
- Demand is unaffected: $MPB = MSB$.



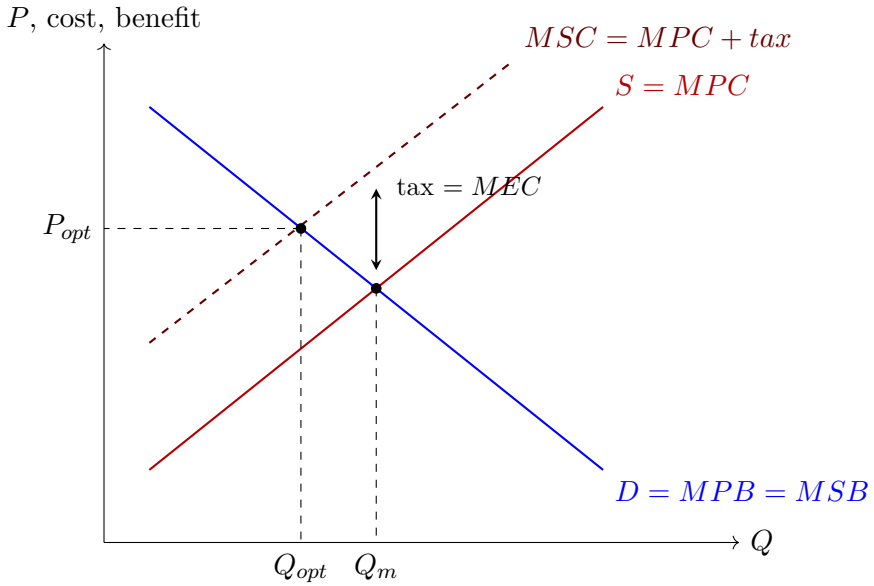
The **free market overproduces**: $Q_m > Q_{opt}$, because producers ignore the external costs. The shaded triangle (WL) is the **welfare loss** — units between Q_{opt} and Q_m have $MSC > MSB$, meaning society would be better off if they were not produced. The vertical gap between MSC and MPC is the **marginal external cost (MEC)** per unit.

$$\text{Welfare loss} = \frac{1}{2}(Q_{\text{m}} - Q_{\text{opt}}) \times (MSC - MPC)$$

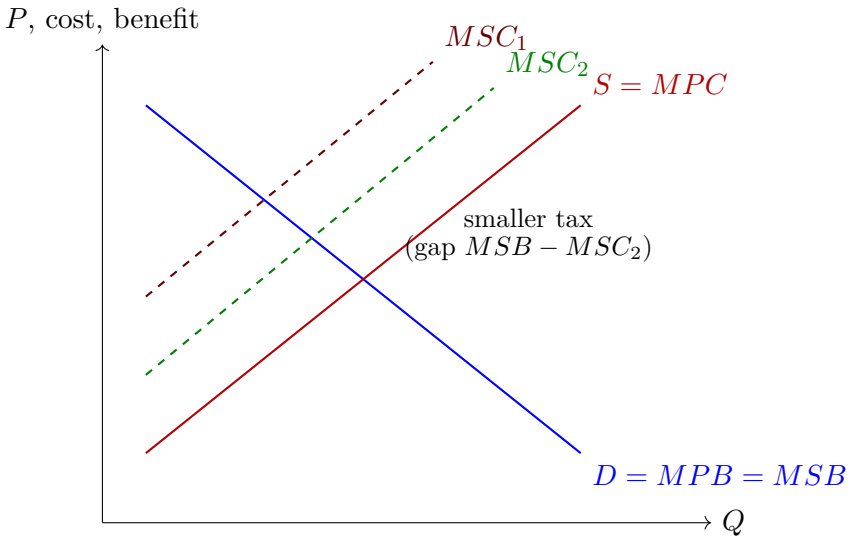
The welfare loss could also be understood as a change in total social benefits (change in CS and PS)

Policies to correct negative production externalities and evaluation

- Market policies
  - Indirect taxes / pigouvian taxes: imposing a tax equal to the **external cost** (MEC) per unit, so the $S = MPC$ curve shifts upward to overlap with MSC. The new equilibrium is at the intersection of MSC and $D = MPB = MSB$, yielding the **optimal** quantity Q_{opt} and price P_{opt}.

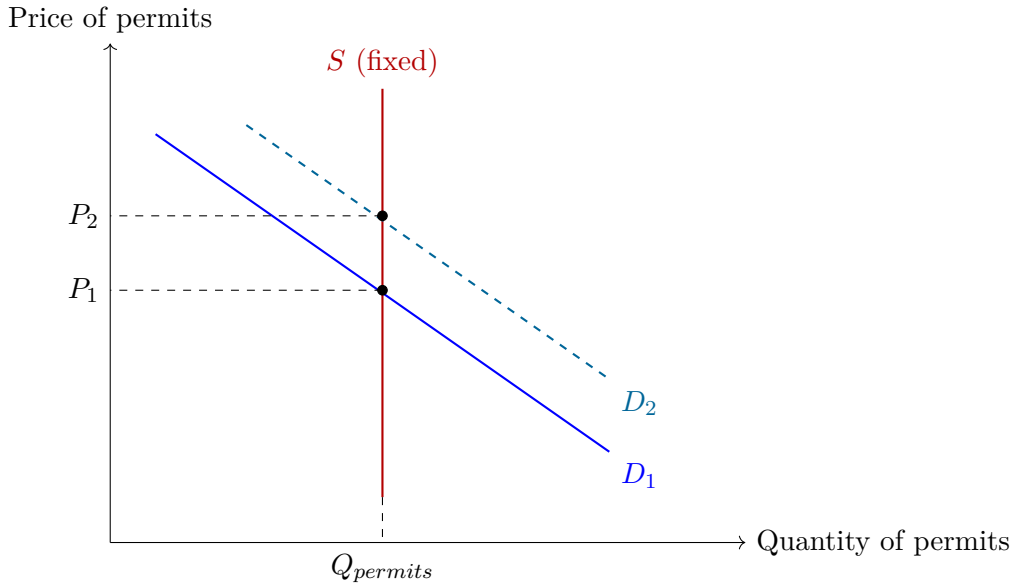


– Carbon taxes: tax per unit of carbon emissions of fossil fuels. Following the imposition of the tax, firms must pay the higher price to buy the fossil fuel. The increase in the price of the high-carbon fuel creates incentives for firms to switch to other, less polluting or non-polluting energy sources. If a firm switches to less polluting resources, the external costs of production fall, so the MSC curve shifts **downward** (from MSC_1 to MSC_2). With lower external costs, the optimal quantity **increases** from $Q_{opt,1}$ to $Q_{opt,2}$, and the optimal tax needed (the gap between MSB and MSC_2) is smaller.



– Tradable permits: policy involving permits to pollute issued to firms by a government or an international body. The total pollution allowed must be **less** than the unregulated level for the scheme to reduce pollution. Permits can be traded between firms.

  * The **supply of permits is perfectly inelastic**, fixed by the government.
  * The **demand for permits** determines the equilibrium price.
  * If a firm can produce with lower emissions than its permits allow, it can **sell** the extras → profit.
  * If a firm needs to emit more than its permits allow, it must **buy** more → cost.
  * As the economy grows and output rises, demand for permits shifts **rightward** ($D_1 \rightarrow D_2$), so the permit price rises ($P_1 \rightarrow P_2$).
  * Like carbon taxes, tradable permits incentivise producers to switch to less polluting resources, reducing the externality and shifting MSC downward toward MPC (as in Figure 5.5(b)).



- **Pros (advantages of market-based policies):**
  - **Internalise the externality**: external costs become paid for by producers/consumers, aligning private with social costs.
  - **Taxes on pollutants > taxes on output**: emission taxes (e.g. carbon taxes) incentivise firms to *economise on polluting inputs* and switch to cleaner technology, whereas output taxes only reduce quantity produced with the same dirty technology.
  - **Least-cost pollution reduction**: firms facing lower clean-up costs cut pollution first; high-cost firms pay the tax or buy permits → society achieves a given emission cut at the *lowest overall cost*.
  - **Tradable permits** similarly reward low-cost abaters (they sell excess permits) and penalise high-cost ones (they must buy).
  - **Government revenue**: taxes generate revenue that can be used for related spending (e.g. clean energy investment).
- **Cons (disadvantages of market-based policies):**
  - **Technical difficulty of measuring harm**: enormous uncertainty over which production methods/pollutants cause what damage, and how to attach a monetary value to the harm.
  - **Taxes**: must equal MEC per unit, which is unknown.
    - What production methods produce pollutants?
    - Which pollutants are harmful?
    - What is the value of the harm?
    - What is the appropriate amount of tax?
    - How will consumers be affected?

    In practice, carbon taxes are usually set *too low* for political reasons, limiting their impact.
  - **Indirect taxes are regressive**: lower-income households pay a higher proportion of their income in tax than higher-income households → **inequitable**.
  - **Firms may keep polluting**: even if they pay the tax, some firms do not reduce emissions.
  - **Tradable permits:** setting the maximum acceptable pollution level requires technical info often unavailable. Too high → ineffective; too low → permits become very costly for firms. This results in permits developed for only a few types of pollutants.
  - **Distribution of permits** may involve political favouritism.
  - In practice, the best achievable outcome is usually only a partial shift of MPC toward MSC and some reduction in the externality; full optimality is unlikely.

    - Requires only that users can **communicate** and that the resource has **clear boundaries**.
  - **Cons:**
    - Requires **effective communication** among users to create and agree on rules — fails if communication is poor or self-organisation costs are too high.
    - Requires **well-defined boundaries** for the resource (e.g. a pasture, lake, or wood).
    - **Ill-suited to global resources** such as the oceans or the global climate, where boundaries are hard to define and users cannot easily communicate or enforce rules (Ostrom herself despaired over the oceans).
    - Requires a supporting **legal system of land/resource rights** to function effectively.

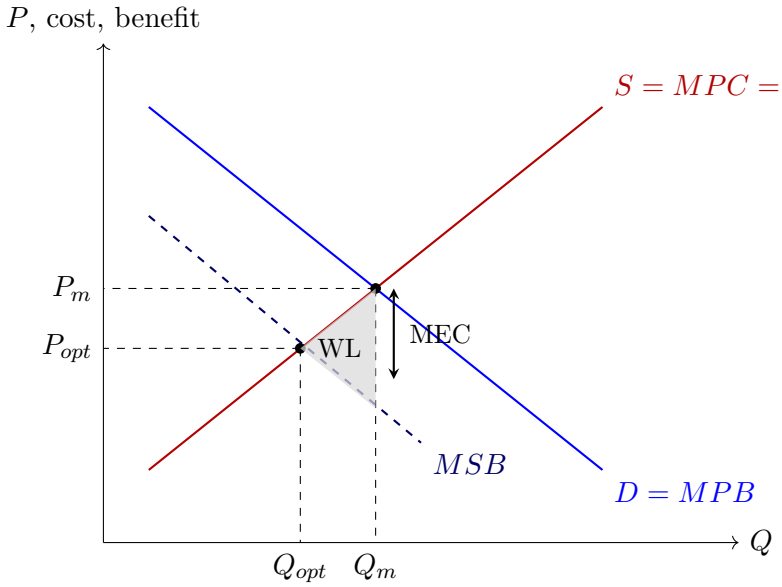
- **Education and awareness creation**: educating the public and providing information about firms' polluting activities can lead consumers to **turn away** from offending firms' products, forcing firms to change their production methods.
  - **Pros**: firms are strongly influenced by customer opinions and want to avoid drops in sales.
  - **Cons**: only makes a **small difference**; effective for localised, high-profile issues (e.g. an oil spill) where consumers may boycott, but broad problems like fossil-fuel-driven climate change require solutions on a far larger scale (carbon taxes, tradable permits are far more effective).

- **International agreements**: negative production externalities and overuse of common pool resources often have **international repercussions**, requiring **cooperation among govts**. Co-operation may be *global* or *regional*, and is also crucial for the development and diffusion of new green technologies. **Key challenge**: global common pool resources (oceans, climate, ozone) are owned by no one, so no single country can claim damages for their destruction — responsibility is diffuse and consequences are felt worldwide, making enforcement difficult.

5.4 Negative consumption externalities

Negative consumption externalities occur when consumption imposes **external costs** on third parties (e.g. second-hand smoke, litter). The social benefit of consumption is less than the private benefit:

- $MSB = MPB - MEB$ (marginal external benefit is negative; i.e. there is a marginal external cost)
- Since there is an external cost, $MSB < MPB \Rightarrow$ the MSB curve lies **below** the MPB (D) curve.
- Supply is unaffected: $MPC = MSC$.



The **free market overconsumes**: $Q_m > Q_{opt}$, because consumers ignore the external costs of their consumption. The shaded triangle (WL) is the **welfare loss** — units between Q_{opt} and Q_m have $MSC > MSB$, meaning society would be better off if they were not consumed. The vertical gap between MPB and MSB is the **marginal external cost (MEC)** per unit.
Welfare loss can be calculated as

$$\text{Welfare loss} = \frac{1}{2}(Q_m - Q_{opt}) \times (MPB - MSB)$$

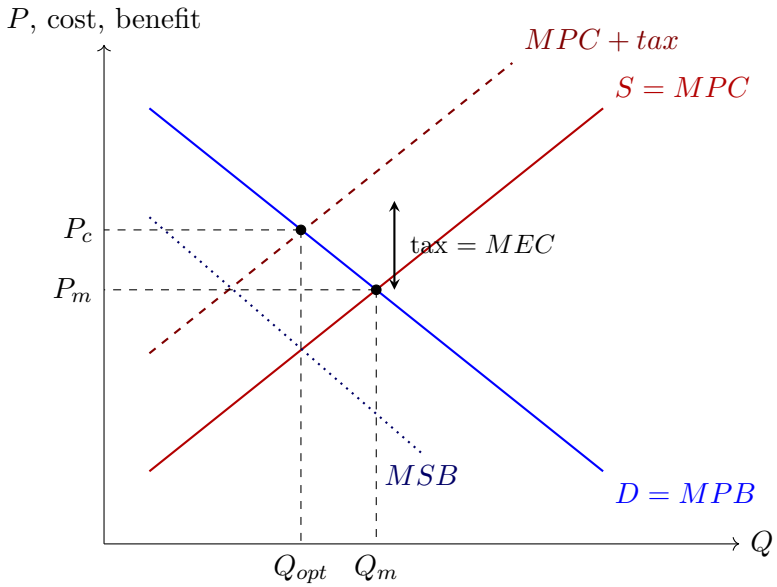
It could also be thought as total social surplus (CS+PS) - total social benefits (CS+PS-external cost)

Demerit goods and overuse of CPR

Demerit goods are goods that are considered to be undesirable for consumers, but which are overprovided by the market. One common reason that they are overprovided is because of negative consumption externalities. Overuse of common pool resources is seen as resulting more from production activities than consumption activities, therefore relating more to negative production rather than consumption externalities. However, overuse of common pool resources also results from negative consumption externalities.

Policies to correct NCE and prevent overuse of CPR

- Market-based
  - **Indirect (Pigouvian) taxes**: imposed on goods whose consumption creates external costs (e.g. cigarettes, petrol/gasoline). The tax shifts the supply curve **upward** from $S = MPC$ to $MPC + tax$. If the tax equals the external cost per unit, the $MPC + tax$ curve intersects $D = MPB$ at the socially optimal quantity Q_{opt}, where allocative efficiency is achieved. The quantity consumed falls from Q_m to Q_{opt}, and the price rises from P_m to P_c. (Note: the demand curve does *not* shift.)



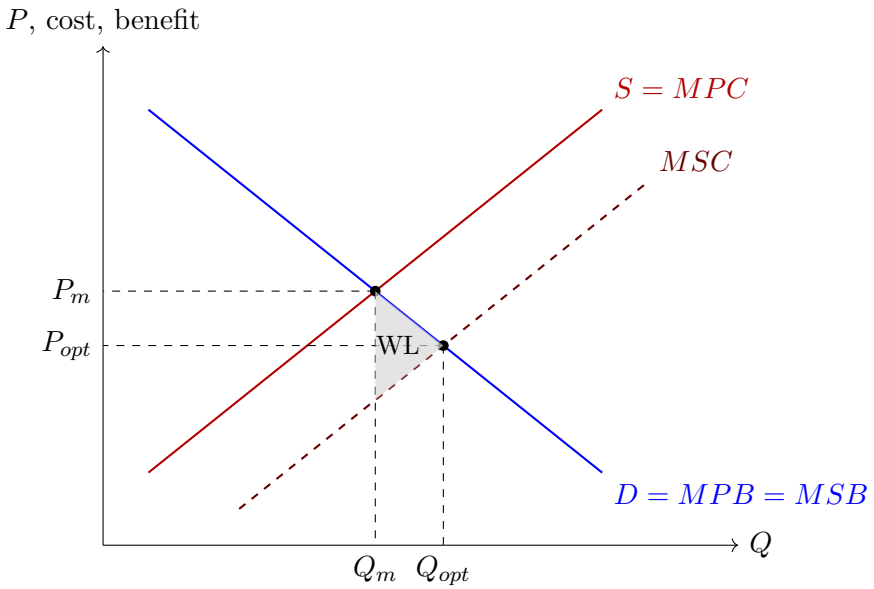
  - **Pros of indirect taxes (market-based policy):**
    - **Internalise the externality**: external costs become paid for by consumers, aligning private with social costs.
    - **Change relative prices**: the taxed good becomes relatively more expensive → consumers reduce consumption.
    - **Preferred by economists** over regulation, as long as the situation permits market-based policy.

6 Market failure and socially undesirable outcomes II

6.1 Positive production externalities

Positive production externalities occur when production creates **external benefits** for third parties (e.g. R&D leading to new technologies, vaccination production). The social cost of production is less than the private cost:

- $MSC = MPC - MEB$ (marginal external benefit treated as a "negative cost")
- Since $MEB > 0$, $MSC < MPC \Rightarrow$ the MSC curve lies **below** the MPC (S) curve.
- Demand is unaffected: $MPB = MSB$.



The **free market underproduces**: $Q_m < Q_{opt}$, because producers ignore the external benefits of their production. At Q_m, $MSB > MSC$, meaning society would benefit from producing more. The shaded triangle (WL) is the **welfare loss** — units between Q_m and Q_{opt} where $MSB > MSC$ are not produced. The vertical gap between MPC and MSC is the **marginal external benefit (MEB)** per unit.

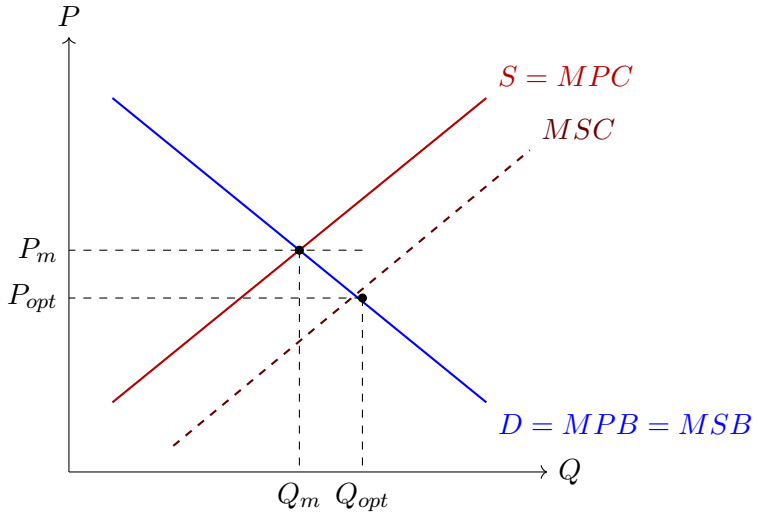
Welfare loss can be calculated as

$$\text{Welfare loss} = \frac{1}{2}(Q_{opt} - Q_m) \times (MPC - MSC)$$

It can also be interpreted by comparing total social benefits at market equilibrium and social optimum.

Correcting positive production externalities

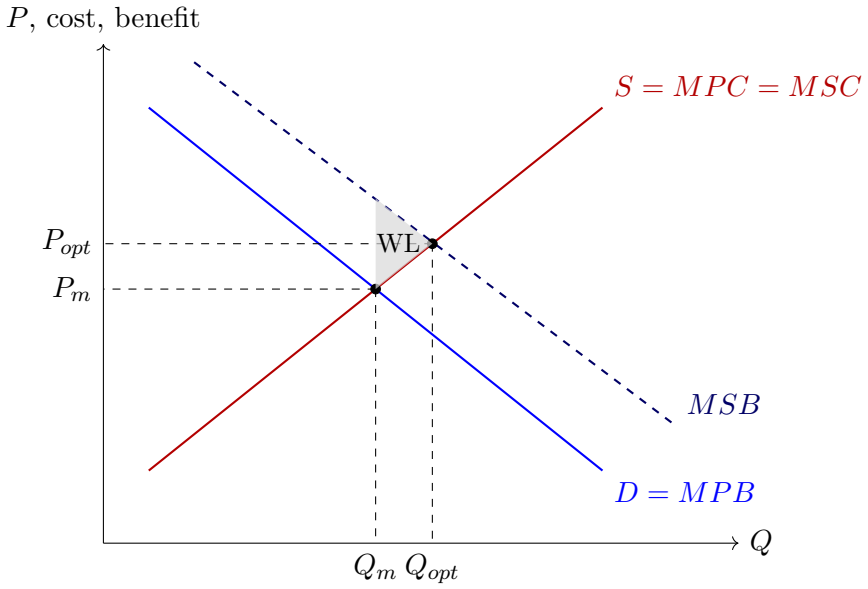
- **Direct government provision**: the government itself provides the good/service creating the positive externality (e.g. R&D for new technology, medicine/pharmaceuticals, worker training), funded by tax revenue. This shifts the $S = MPC$ curve **downward (rightward)** toward the MSC curve, increasing quantity to Q_{opt} and lowering price from P_m to P_{opt}.
- **Subsidies**: the government grants a per-unit subsidy to producers equal to the **external benefit** (MEB). This shifts the MPC curve **downward (rightward)** until it coincides with MSC, increasing output to Q_{opt} and lowering price to P_{opt}. Allocative efficiency is achieved.



6.2 Positive consumption externalities

Positive consumption externalities occur when consumption creates **external benefits** for third parties (e.g. vaccination, education). The social benefit of consumption exceeds the private benefit:

- $MSB = MPB + MEB$ (marginal external benefit)
- Since $MEB > 0$, $MSB > MPB \Rightarrow$ the MSB curve lies **above** the MPB (D) curve.
- Supply is unaffected: $MPC = MSC$.



The **free market underconsumes**: $Q_m < Q_{opt}$, because consumers ignore the external benefits of their consumption. At Q_m, $MSB > MSC$, meaning society would benefit from consuming more. The shaded triangle (WL) is the **welfare loss** — units between Q_m and Q_{opt} where $MSB > MSC$ are not consumed. The vertical gap between MPB and MSB is the **marginal external benefit (MEB)** per unit.

Welfare loss can be calculated as

$$\text{Welfare loss} = \frac{1}{2}(Q_{opt} - Q_m) \times (MSB - MPB)$$

It could also be interpreted as the difference between social benefits at social optimum - market equilibrium. (CS+PS)

7.2 Profit maximisation by the rational producer

Revenue

Revenues are the payments firms receive when they sell the goods and services they produce.

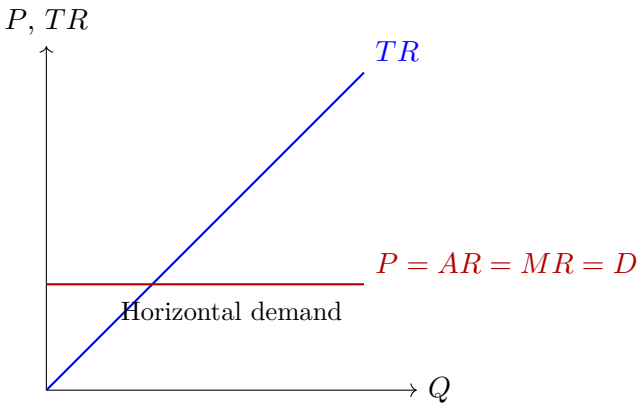
- Total revenue, $TR = Q \times P$ maximized where $PED = 1$ and $MR = 0$.
- Average revenue, $AR = \dfrac{TR}{Q} = P$
- Marginal revenue, $MR = \dfrac{\Delta TR}{\Delta Q}$

Revenues in two different types of markets:

- Perfect competition: the firm is unable to control price; price is constant as output varies
- Monopolistic competition, Oligopoly, Monopoly: the firm has control over price; price varies with output

Perfect competition: revenue data (price is constant; $P = AR = MR$):

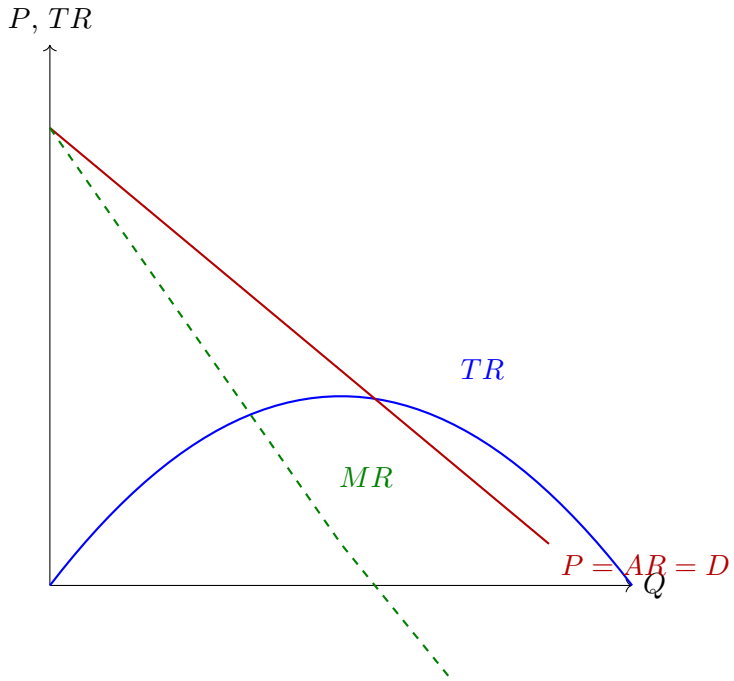
Q	P	TR	AR	MR
0	10	0	10	10
1	10	10	10	10
2	10	20	10	10
3	10	30	10	10
4	10	40	10	10
5	10	50	10	10
6	10	60	10	10



In perfect competition, $P = AR = MR$ is a **horizontal line** (the firm is a price taker), and TR is an **upward-sloping straight line** through the origin ($TR = P \times Q$).

Imperfect competition (monopoly, monopolistic competition, oligopoly): revenue data (firm faces downward-sloping demand; $P = AR > MR$):

Q	$P = AR$	TR	AR	MR
0	11	0	—	—
1	10	10	10	10
2	9	18	9	8
3	8	24	8	6
4	7	28	7	4
5	6	30	6	2
6	5	30	5	0
7	4	28	4	−2



Costs of production

When firms use inputs or resources to produce, they incur CoP which include money payments to buy resources plus anything else given up by a firm for the use of resources.

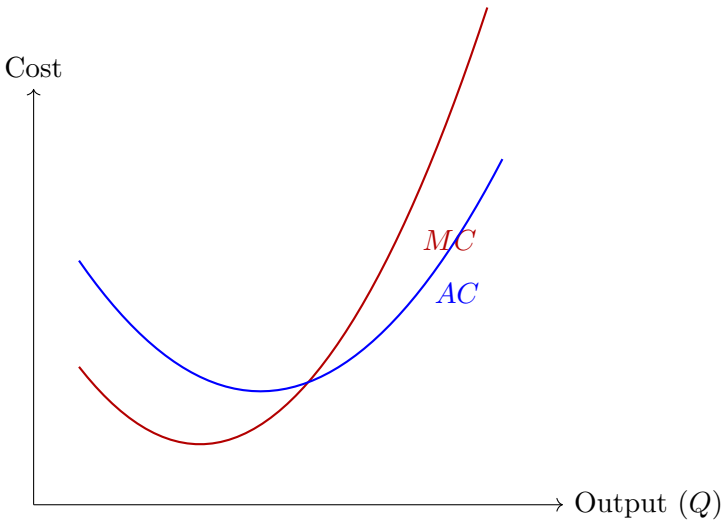
- Such payments made by a firm to outsiders to acquire resources for use in production are known as **explicit costs**.
- The sacrificed income arising from the use of self-owned resources by a firm is an **implicit cost**. (Considered as OPC?)

The sum of explicit and implicit costs incurred by a firm for its use of resources, whether purchased or self-owned, are known as **economic costs**.
Different types of cost:

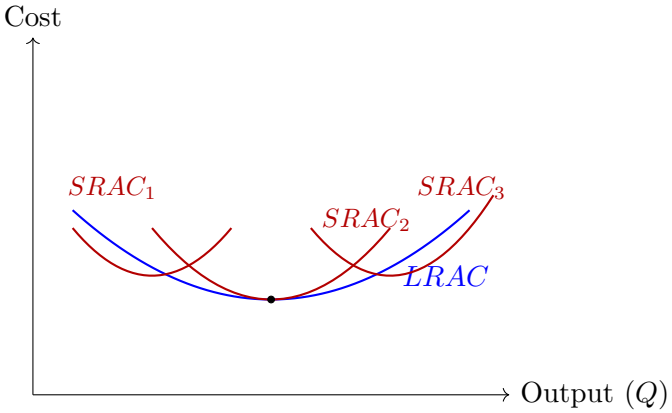
- Total cost, TC, all the CoP incurred by a firm
- Marginal cost, $MC = \dfrac{\Delta TC}{\Delta Q}$
- Average cost, $AC = \dfrac{TC}{Q}$

Cost of production under short run and long run:

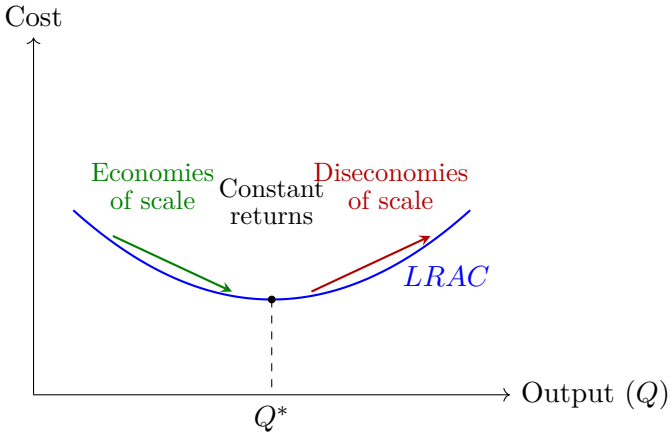
- In the short run: the period of time when at least one factor of production is fixed
  Key relationship between MC and AC:
  - When $MC < AC$, AC is **falling** (the extra unit costs less than the average, pulling the average down).
  - When $MC > AC$, AC is **rising** (the extra unit costs more than the average, pulling the average up).
  - When $MC = AC$, AC is at its **minimum** (MC cuts AC at its lowest point).
  - The same relationship holds between MC and AVC (average variable cost).



- In the long run: the period of time when all factors of production are variable (increases all FoPs)



(a) SRAC curves trace out LRAC



(b) EOS and DOS

Economies of scale: decreases in average costs of production over the long run as a firm increases all its factors of production.
Reasons for EoS to occur:

- Specialization of labor (labor economies)
- Specialization of management (managerial economies)
- Bulk buying of inputs / FoPs (buying economies)
- Financing economies
- Spreading of certain costs, such as marketing, over larger volumes of output. (risk-bearing economies)

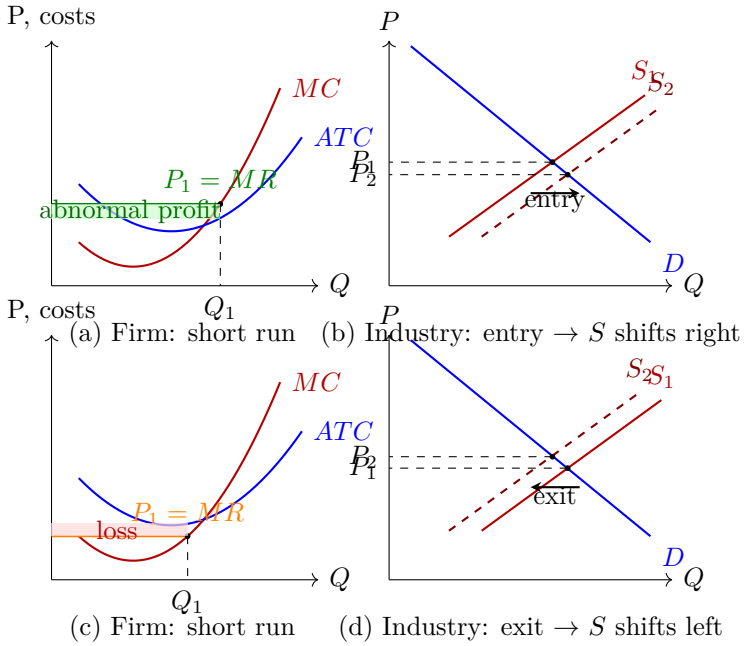
Diseconomies of scale: increases in the average costs of production in the long run as a firm increases its output by increasing all its inputs.
Reasons for DisEoS to occur:

- Co-ordination and monitoring difficulties
- Communication difficulties
- Poor worker motivation

4. Entry continues until abnormal profit is competed away: at P_2, every firm earns **normal profit** where P = minimum ATC.

Losses in the short run → normal profit in the long run:

1. In the short run, price P_1 < minimum ATC, so firms make losses (loss per unit = $a - b$ in the textbook figure).
2. In the long run, firms are free to leave the industry; as some **exit**, industry supply S_1 shifts **left** to S_2.
3. Price rises from P_1 to P_2; remaining firms' losses shrink until, at P_2, they earn **normal profit** (P = minimum ATC).



(a) Firm: short run (b) Industry: entry → S shifts right

(c) Firm: short run (d) Industry: exit → S shifts left

Both adjustment processes are driven by the **MC = MR** rule at every stage.

Allocative efficiency

Allocative efficiency occurs when firms produce the particular combination of goods and services that consumers mostly prefer. Allocative efficiency is achieved when MB = MC; but since MB = P, it follows there is allocative efficiency when P = MC. Note that this condition holds only when there are no externalities, in which case it is also true that MSB = MSC.

- If P > MC, underallocation of resources devoted for production
- If P < MC, overallocation of resources devoted for production

In long-run equilibrium under perfect competition, the firm achieves allocative efficiency where P = MC (or where MB = MC). At the level of the industry, social surplus (consumer plus producer surplus) is maximum, and MB = MC.

- refusing to deal with certain customers or offering special discounts to customers who buy all or most of their supplies from the dominant company
- making the sale of one product conditional on the sale of another product

Legislation to protect competition (competition policy)

Most countries have laws to **prevent collusion** among oligopolists and **anti-competitive behaviour** by dominant firms. Firms found guilty face **fines** or may be **broken up** into smaller firms. Difficulties:

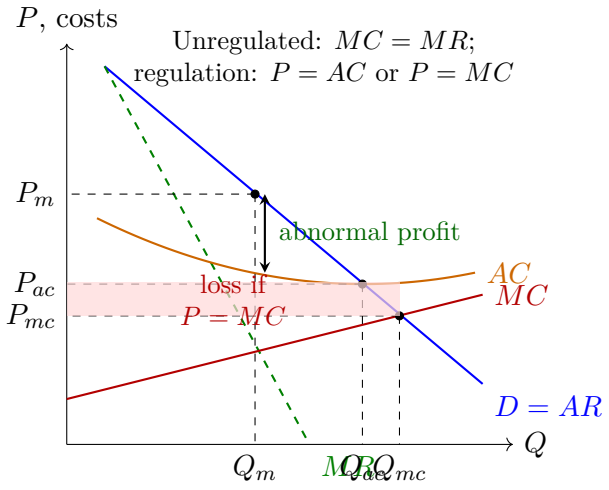
- **Interpretation**: laws may be vague; distinguishing abuse of market power from legitimate action is highly controversial.
- **Enforcement varies** by government priorities/ideology — some believe strict intervention is needed; others believe market forces self-correct over time (a normative issue).
- **Collusion is secret** and hard to discover/prove.

Mergers: a merger is an agreement by two or more firms to combine into one. Mergers may pursue EOS, growth, or market power. Competition policy limits mergers that would create too much market power. *Difficulties*: uncertainties over which mergers to allow; ideological differences on acceptable market power. **Fines**: firms may calculate that profits from illegal behaviour exceed the fine (especially if they might not get caught). Big firms may ignore ethics if non-compliance is more profitable than compliance.

Regulation of natural monopolies

Natural monopolies are not broken up (doing so would raise AC and waste resources). Two main regulatory approaches:

- **Marginal cost pricing** ($P = MC$): achieves **allocative efficiency** (lowest P, highest Q). However, for a natural monopolist $P = MC < AC$, so the firm makes **losses** → requires government subsidy or the firm shuts down. Not a popular method.
- **Average cost pricing** ($P = AC$): the firm earns **normal profit** (no loss, no shutdown). Results in higher P and lower Q than marginal cost pricing, but still **better than the unregulated monopoly outcome**. Allocative efficiency is *not* achieved but welfare loss is reduced.
  Disadvantages: removes the incentive to keep costs low (if costs rise, the regulated price rises too); the firm may continue to be protected even if it is no longer a natural monopoly due to technological change (e.g. telecommunications).



Government ownership: an alternative where the government owns the natural monopoly and can force lower prices / higher output. *Disadvantage*: government firms lack profit-maximising incentives → inefficiency and higher costs. Global trend since the 1990s has been toward **privatization**.

7.8 Supplementary: Factor (resource) markets and monopsony (AP Barron Ch 10 & Edexcel Book 2 Ch 13–14)

In factor (resource) markets, firms are the **buyers** of factors of production (such as labour, capital, and land) and households are the **sellers**.

- **Derived demand**: demand for a factor of production is not demanded for its own sake, but is **derived** from the demand for the final goods and services it helps produce.
- **Marginal revenue product (MRP)**: the additional revenue generated by employing one more unit of an input:

$$MRP = MP \times P \qquad \text{or} \qquad MRP = \frac{\Delta TR}{\Delta L}$$

  Because of the law of diminishing returns, MP eventually falls, making the MRP curve downward-sloping. The MRP curve represents the firm's **demand curve for labour**.
- **Marginal factor cost (MFC)**: the additional cost incurred by employing one more unit of an input:

$$MFC = \frac{\Delta TC}{\Delta L}$$

- **Profit-maximising hiring rule**: a firm hires labour up to the point where:

$$MRP = MFC$$

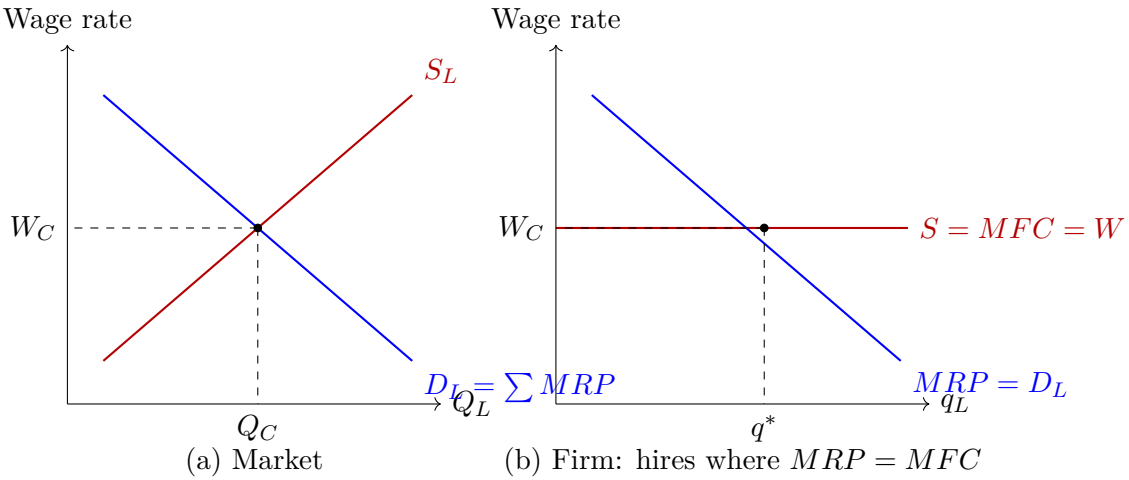
- In a **perfectly competitive labour market**, each individual firm is a **wage taker** facing a horizontal labour supply curve at market wage W. Therefore $MFC = W$, and the firm hires where $MRP = W$.
- **Least-cost rule** for two inputs (labour L and capital K):

$$\frac{MP_L}{P_L} = \frac{MP_K}{P_K}$$

  To maximise total profits, the firm also ensures that:

$$\frac{MRP_L}{P_L} = \frac{MRP_K}{P_K} = 1$$

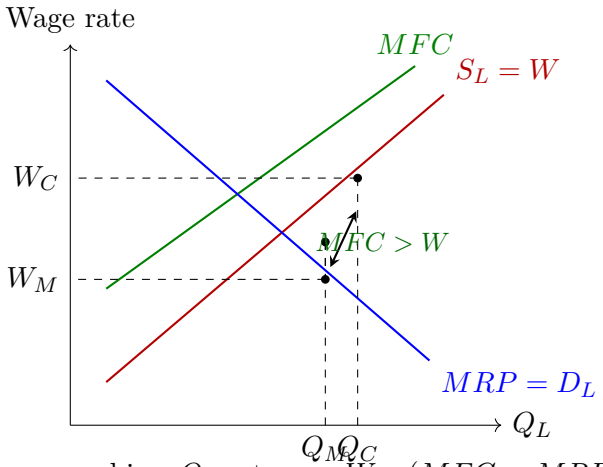
Perfectly competitive labour market (many firms hiring workers with similar skills; each firm is a **wage taker**):



(a) Market (b) Firm: hires where $MRP = MFC$

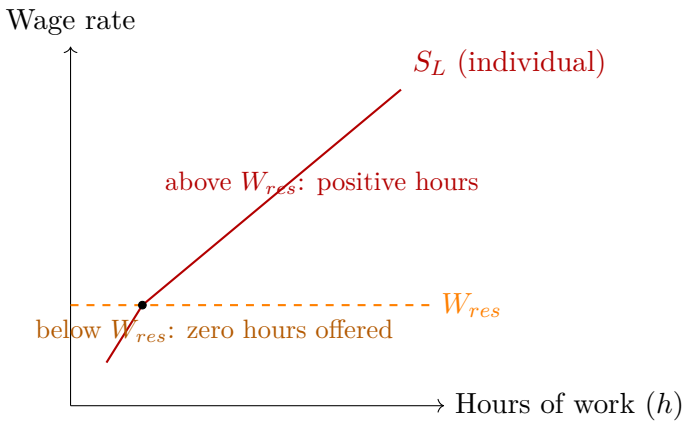
Monopsony (single buyer of labour, e.g. the sole dominant employer in a town or an industry buyer):

- Because the monopsonist faces the entire **upward-sloping market supply of labour**, hiring an additional worker requires paying a higher wage not only to the marginal worker but to **all previously hired workers**.
- Consequently, the marginal factor cost lies strictly **above** the labour supply curve: $MFC > W$.
- The monopsonist profit-maximises by hiring where $MFC = MRP$ (quantity Q_M), but pays the lower wage W_M determined by the supply curve at Q_M.
- Outcome: the monopsonist hires **fewer workers at a lower wage** ($Q_M < Q_C$ and $W_M < W_C$) compared to a competitive market, creating deadweight loss and market failure.
- **Policy remedy**: setting a **minimum wage** between W_M and W_C makes the labour supply curve horizontal up to that minimum wage, eliminating the upward-sloping MFC penalty. Under monopsony, a minimum wage can therefore **increase both wages and employment** toward the competitive level.



Monopsony hires Q_M at wage W_M ($MFC = MRP$); a competitive market would hire Q_C at W_C

– **Labour income taxes**: taxes reduce net take-home pay, effectively requiring a higher gross wage to meet the worker's reservation threshold. Targeted tax credits (e.g. Earned Income Tax Credit) raise net pay and boost labour supply.



10.2 Low and stable rate of inflation

Inflation: sustained rise in general price level (an average of prices of goods and services in the entire economy).
Deflation: a sustained decrease in the general price level. **Disinflation**: a decrease in the rate of inflation. Measures of inflation and deflation are obtained by the use of price indices (CPI, PPI, RPI...), measure of average prices in one period relative to average prices in a reference period called a base period.
The **consumer price index (CPI)** is a measure of the cost of living for the typical household, and compares the value of a basket of goods and services in one year with the value of the same basket in a base year. Inflation (and deflation) are measured as a percentage change in the value of the basket from one year to another.

$$PI = \frac{\text{Value of basket in a specific year}}{\text{Value of the same basket in the base year}} \times 100$$

(Remember to consider weights for constructing PI if there exists such weights)

$$\Delta\%\text{Inflation / Deflation} = \frac{PI_2 - PI_1}{PI_1} \times 100$$

$$\text{real income} = \frac{\text{nominal income}}{CPI} \times 100$$

Problems with CPI:

- **Different rates of inflation for different income earners**: the CPI basket reflects the *average* household, but consumption patterns vary by income level → different groups face different inflation rates.
- **Different rates of inflation by regional/cultural factors**: tastes and purchases vary by region and culture, so the CPI may not reflect specific groups' actual inflation.
- **Consumer substitutions when relative prices change**: as some goods become cheaper/more expensive, consumers substitute, but fixed weights in the basket cannot capture this → CPI usually **overstates** inflation.
- **Increasing use of discount stores and sales**: consumers buy at lower prices than those used in CPI calculations → CPI overstates inflation.
- **Introduction of new products**: a fixed basket cannot account for new products (e.g. Netflix replacing DVDs) or products becoming obsolete.

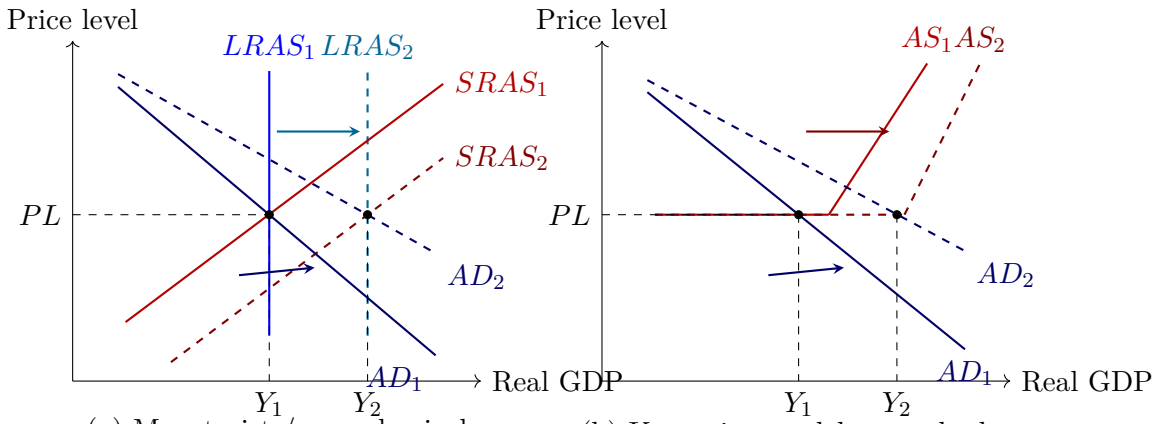
Impact of economic growth on unemployment and inflation (supplementary)

Growth and unemployment:

- **Short-term growth** (expansionary phase of the business cycle) reduces **cyclical unemployment**. If AD keeps rising beyond full employment (inflationary gap), unemployment can even fall **below** the natural rate — but only temporarily, since contractionary policies (or self-correction) bring it back.
- **Long-term growth** (increases in potential output, rightward shifts of LRAS/Keynesian AS) can reduce **natural** and especially **structural** unemployment (via supply-side factors: more/better resources, technology). But **not all growth lowers unemployment**: growth driven by technological change may *increase* structural unemployment (skills become obsolete), and capital-intensive technologies in developing countries may raise unemployment.

Growth and inflation:

- In the **Keynesian** model, growth from AD increases is non-inflationary while there is spare capacity (horizontal AS); as real GDP approaches potential output, resource bottlenecks push prices up, and beyond potential output further AD growth is highly inflationary.
- In the **monetarist/new classical** model, an AD increase *always* raises the price level in the short run, and in the long run only raises the price level.
- Both models agree: **long-term growth** (increases in potential output) **reduces inflationary pressure** — AD growth matched by LRAS/AS growth does not raise prices.
- The relationship also runs the other way: unemployment and inflation affect growth (unemployment wastes resources; high inflation discourages investment and net exports).



(a) Monetarist / new classical growth with no inflation

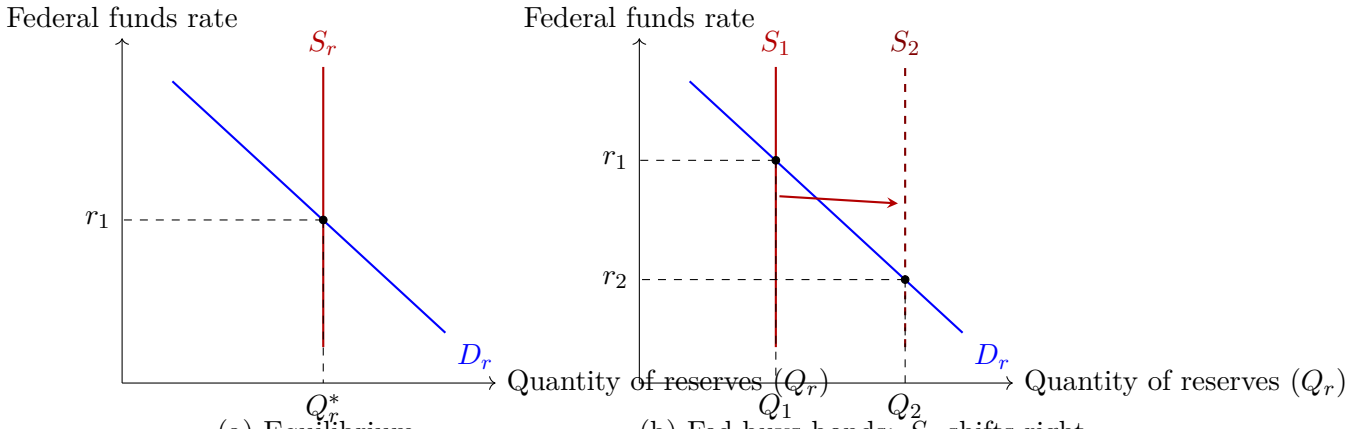
(b) Keynesian model: growth along the horizontal AS with no inflation

11.2 Sustainable level of government debt

Government / national / public debt: the amount of money that a government owes to lenders outside of the government itself.

- If expenditures are larger than tax revenues, there is a budget deficit; solved through borrowing money from the private sector.
- If expenditures are smaller than tax revenues, there is a budget surplus.

- **Contractionary policy**: the central bank **sells government securities** → deducts reserves from buyer banks' accounts → S_r shifts **left** → the federal funds rate **rises**, raising commercial borrowing rates and shifting AD leftward.
- **Administered rates**: the **discount rate** (lending rate directly from the central bank) acts as an upper ceiling for the market rate, while interest paid on reserve balances (IORB) sets a lower floor.



(a) Equilibrium (b) Fed buys bonds: S_r shifts right

Nominal IR: the market rate that prevails at any moment in time.
Real IR: the interest rate that has been corrected for inflation, = Nominal IR - Rate of inflation.

Supplementary: Quantity theory of money, loanable funds and the Fisher effect (AP Barron Ch 15)

- **Equation of exchange / quantity theory of money**:

$$MV = PY$$

where M = money supply, V = velocity of money (how many times a unit of currency changes hands), P = price level, and Y = real GDP (PY = nominal GDP).

  - Classical and monetarist assumption: V and real output Y are determined by real factors and are relatively stable in the long run.
  - Therefore, any increase in money supply (M) translates proportionally into a higher price level (P) — "inflation is always and everywhere a monetary phenomenon".

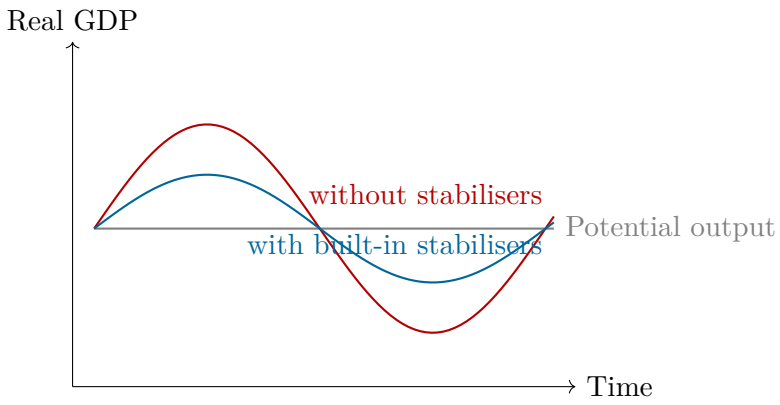
- **Loanable funds market**: a model analysing the real interest rate determined by the demand for and supply of loanable funds:
  - **Demand for loanable funds**: originates from firms undertaking **investment** expenditure and government borrowing to finance deficits (downward-sloping with respect to the real interest rate).
  - **Supply of loanable funds**: originates from household and business **savings** plus foreign capital inflows (upward-sloping with respect to the real interest rate).
  - **Crowding out**: government budget deficits increase the demand for loanable funds (shifting demand right), which pushes up the real interest rate and crowds out private investment.

- **Fisher effect**: the nominal interest rate equals the real interest rate plus the expected inflation rate:

$$\text{Nominal interest rate} \approx \text{Real interest rate} + \text{Expected inflation}$$

In the long run, changes in expected inflation lead to one-for-one adjustments in nominal interest rates, leaving the equilibrium real interest rate determined by savings and investment fundamentals.

- **What they can and cannot do**:
  - They **dampen** fluctuations: they can prevent a recession from becoming a depression and help prevent an expansion from turning into hyperinflation.
  - They **cannot prevent a recession or close an output gap on their own** — a drop in income is what activates them, and they only reduce the size of the swing.
  - They work through the **multiplier**: by reducing the size of the **MPC** (higher taxes and benefits leak more of each income change out of the spending stream), they lower the multiplier, so any given shock to AD produces smaller induced changes in real GDP (e.g. if unemployment benefits lower the MPC from 3/4 to 1/2, the multiplier falls from 4 to 2).



Built-in stabilisers reduce the amplitude of the business cycle around potential output

Evaluation: built-in stabilisers are **fast** (no recognition/decision/implementation lags), **automatic** (no political battle), and **self-reversing** (they do not add permanently to debt — deficits in recessions are offset by surpluses in booms). Their main limitation is that they are **only partially stabilising**: they cushion shocks but cannot pull an economy out of a deep recession, which is why governments still use **discretionary** fiscal policy for large gaps. Developing countries often have **weak built-in stabilisers** because welfare spending is a small share of their budgets. **Monetary vs. fiscal policy for demand management**: most economists favour **monetary policy** for routine demand management (incremental, flexible, reversible, shorter lags, less political interference). **Fiscal policy** plays a crucial role in **deep recessions** where monetary policy may become ineffective.

13.4 The Keynesian multiplier

Defined as

$$\text{Multiplier} = \frac{\Delta\text{Real GDP}}{\text{Initial change in expenditure}}$$

The change in real GDP is likely to be greater than the initial change in expenditure (the initial change in expenditure causes a chain reaction of further expenditures, called *induced spending*: increased investment means money is paid to the owners of FoPs, who then use it for increased consumption, causing GDP to increase further), causing the value of the multiplier to be >1.

- MPC, marginal propensity to consume: fraction of additional income households spend on domestically produced G&S.
- MPS, marginal propensity to save: fraction of additional income saved.
- MPT, marginal propensity to tax: fraction of additional income taxed
- MPM, marginal propensity to import: fraction of additional income spent on imported goods and services

An increase in G enters directly into aggregate demand in its entirety before generating secondary rounds of induced consumption.

- **Tax multiplier**:

$$k_T = -\frac{MPC}{1 - MPC}$$

The tax multiplier is **negative** and **smaller in magnitude** than the spending multiplier. A tax cut of \$1 first increases disposable income by \$1, but households save a fraction (MPS) and spend only $MPC \times \$1$, meaning the first round of spending is smaller.

- **Balanced-budget multiplier**:

$$k_{BB} = k_G + k_T = \frac{1}{1 - MPC} + \left(-\frac{MPC}{1 - MPC}\right) = \frac{1 - MPC}{1 - MPC} = 1$$

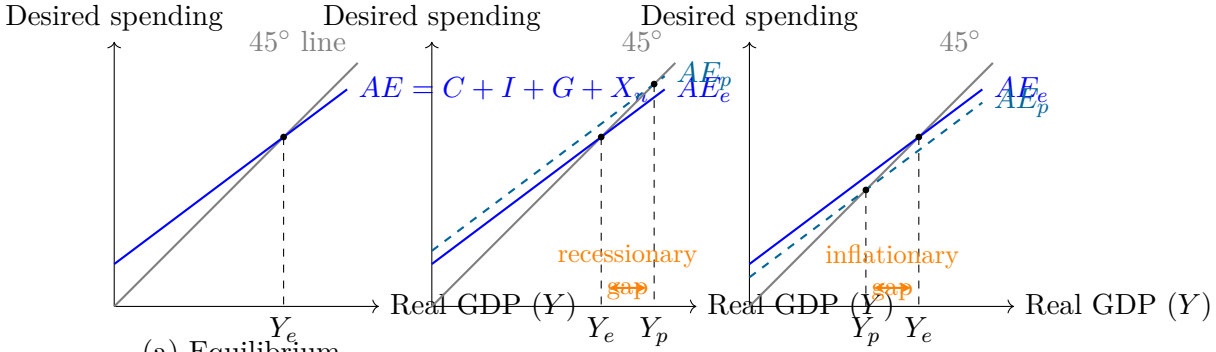
When the government increases government spending and taxes by the exact same amount ($\Delta G = \Delta T$), real GDP expands by exactly the amount of the spending increase (multiplier equals 1).

The Keynesian cross model

Jelps understand the relationship between AD and (real GDP), the idea of **less-than-full-employment equilibrium**, and the **multiplier effect**.
Key distinction: **aggregate output (real GDP)** = sum of all *actual* expenditures ($C + I + G + X - M$). **Aggregate demand** = sum of all *desired* expenditures ($C + I + G + X - M$). The two are not always equal.
Consumption function: desired consumption (C) depends positively on income (Y). The slope of the C function = $\Delta C / \Delta Y$ = the **MPC**. At low income, $C > Y$ (negative saving / dissaving); at high income, $C < Y$ (positive saving).
Adding investment: desired investment (I) is **autonomous** (independent of income). The $C + I$ line is parallel to the C line (shifted up by a constant amount).
Equilibrium: occurs where the **desired spending line ($C + I$)** intersects the **45° line** (where desired spending = income = real GDP). At this point, the amount buyers want to buy equals the amount produced → equilibrium output Y_e.

- If output $< Y_e$: desired spending > output → inventories sold → firms increase production → GDP rises toward Y_e.
- If output $> Y_e$: desired spending < output → unsold goods accumulate → firms cut production → GDP falls toward Y_e.

Crucially, Y_e may be **below full-employment output** → the economy can be in equilibrium with **cyclical unemployment** (a deflationary gap). This was Keynes's key insight against the classical view that the economy self-corrects to full employment.



(a) Equilibrium (b) $Y_e < Y_p$: recessionary (deflationary) gap (c) $Y_e > Y_p$: inflationary gap

Bindles: good produced by many countries around the world, allowed for free trade.
The world bindle price is determined by world demand and world supply, where world demand is the sum of all country demands and world supply is the sum of all country supplies.

Whether the goods should be exported / imported depends on the domestic price relative to the world price.

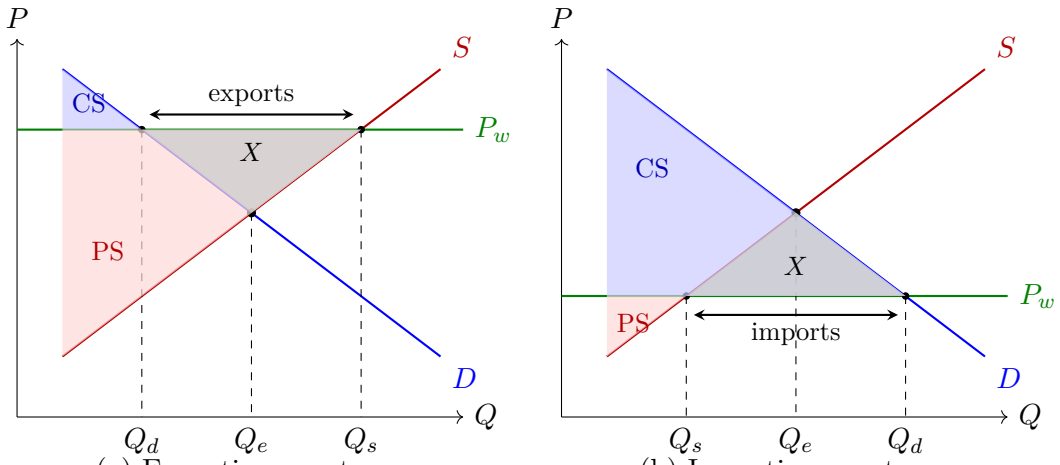
- **Domestic price > world price**: domestic producers cannot compete with cheaper foreign goods → imports are encouraged; domestic consumers benefit from lower prices.
- **Domestic price < world price**: domestic producers can sell at higher prices abroad → exports are encouraged; domestic producers benefit from higher prices.

Note that the world supply curve is perfectly elastic.

Calculation

Quantities exported / imported is determined by the difference between domestic supply and domestic demand at the world price.
Export revenue / import expenditure = world price × quantity exported / imported.
For consumers, they are worse off in case of exports and better off in case of imports.
For producers, they are better off in case of exports and worse off in case of imports.
Effects on surplus (supplementary, recommended for HL): with trade, the changes in producer surplus (PS), consumer surplus (CS) and social surplus can be calculated from the domestic demand and supply curves at the world price P_w:

- **Exporting country**: producer revenue rises (larger quantity at the higher world price) → **PS rises**; consumers pay more and buy less → **CS falls**. The gain in PS **exceeds** the loss in CS by triangle X (between the world price and the autarky price, over the extra exported quantity), so **social surplus rises**.
- **Importing country**: consumers pay less and buy more → **CS rises**; producers receive less and sell less → **PS falls**. The gain in CS **exceeds** the loss in PS by triangle X, so **social surplus rises**.



(a) Exporting country
$P_w > P_e$: PS rises by more than CS falls

(b) Importing country
$P_w < P_e$: CS rises by more than PS falls

In both cases, trade raises social welfare relative to autarky; the only difference is **who gains**: with exports, producers gain at the expense of consumers; with imports, consumers gain at the expense of producers.